

GENERATIVE AI IN VIRTUAL REALITY FOR
MICROBIOLOGY:
A USER STUDY ON ENGAGEMENT AND LEARNING
EFFECTIVENESS

by

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Abstract

This study explored the integration of generative Artificial Intelligence (AI) with Virtual Reality (VR) in enhancing microbiology education through BioVRium, the platform developed for this project. Current research in educational technology has demonstrated that immersive and interactive environments can significantly improve learning outcomes. However, the specific impact of generative AI in VR settings, particularly in the context of complex scientific subjects like microbiology, has been less explored. This research focused on the possible benefits of learning outcomes and usability levels when combining these novel technologies.

To gather user requirements for BioVRium, a 1-week study was conducted through a verified Discord channel designed for research related to Virtual Reality (VR). Quantitative and qualitative data have been collected and analysed to ensure the design of a positive user experience and an efficient learning platform. The participants' (n=31, age= 25-55, SD = 12.33) preferences were pivotal for developing BioVRium, which was developed using Unreal Engine 5.3.2 and the CONVAI plugin. CONVAI was used to create 3 characters – Anca, Dr.Ellis and BioBear. Their role was to guide the user through the platform and be a source of knowledge, representing a virtual professor. Each one of these characters has a distinctive personality; the rationale was to assess if the presence of an AI guide was beneficial for the platform's usability and if it could improve users' retention and motivation. Furthermore, this research examined the participants' preferences for AI guides and the reasons behind these preferences.

In the second phase of this research, a crossover study was employed, involving participants with an advanced academic background (n=10, mean age = 28, SD = 8.11, 5 females/5 males). They interacted with both the VR-only and the AI-enhanced versions of BioVRium. Each condition (control and AI) was tested 5 times with each topic (bacteria and viruses), and each condition-topic combination was presented first and second an equal number of times. The study assessed various factors including user engagement, learning effectiveness, system usability and AI preferences. Key metrics evaluated were task completion rates, error rates, and overall user satisfaction.

The findings were statistically significant. This indicated that integrating generative AI in VR can enhance user engagement and learning outcomes, with participants showing improved performance and higher satisfaction rates in the AI-enhanced sessions compared to the VR-only sessions.

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List of Abbreviations

1. **AI** Artificial Intelligence
2. **AIIS** Artificial Intelligence, Innovation, and Society
3. **API** Application Programming Interface
4. **AR** Augmented Reality
5. **HEI** Higher Education Institutions
6. **NLP** Natural Language Processing
7. **NPC** Non-Playable Character
8. **SD** Standard Deviation
9. **STEM** Science, Technology, Engineering, and Mathematics
10. **SUS** System Usability Scale
11. **UI** User Interface
12. **UX** User Experience
13. **VR** Virtual Reality

1 Literature Review

1.1 Introduction

Microbiology plays a pivotal role in the educational journey of students. It not only elucidates the fundamental principles of molecular life but also emphasizes the significance of hands-on experience in this field. A deep understanding of cellular structures, their functions, and various biological processes is vital. It allows students to appreciate how these microscopic entities contribute to broader physiological systems and how they adapt to pathological conditions. This comprehension is crucial for fostering a holistic view of biological sciences (Priya Prakash Budde, 2012). Ana Laura Sandoval Mantovani (2019) argues that the importance of this subject goes beyond its direct relevance to clinical medicine; as she explained, it serves as a foundation for studies in related fields. Despite its relevance, typical pedagogical techniques for teaching cellular biology have significant drawbacks. Textbooks or two-dimensional images do not capture the intricate biological processes or their three-dimensional character. On top of that, microscopy is not widely available and can only offer a restricted viewpoint.

These difficulties highlight the need for an innovative tool that can overcome the limitations of conventional education (Bermejo, 2023). An excellent example of innovation in response to these educational difficulties is technology-assisted learning. With the introduction of interactive software and multimedia tools, technology has opened new possibilities for learning (Henry Matovu, 2022). Virtual Reality is one of the most exciting technical developments of the last few years and not only for the gaming industry; it has shown significant promise in boosting students' knowledge of difficult subjects. In addition to accommodating different learning styles, this interactive method of instruction also meets the needs of today's students who prefer interactive, digital learning resources (Henry Matovu, 2022).

Studies have shown that VR-assisted learning can raise motivation, engagement, and academic achievement in a variety of fields (see Fig.1), thus these results highlight the need for more research on the topic (Vola, 2023). Apart from its immersive nature, VR helps students by offering the possibility to engage with 3D models. Incorporating 3D models of cellular components and processes complements the spatial and interactive aspects of the subject matter (Henry Matovu, 2022). This incorporation of VR into medical curricula shows great potential for bringing the abstract to life, creating a learning experience that is both captivating and impactful (Henry Matovu, 2022).

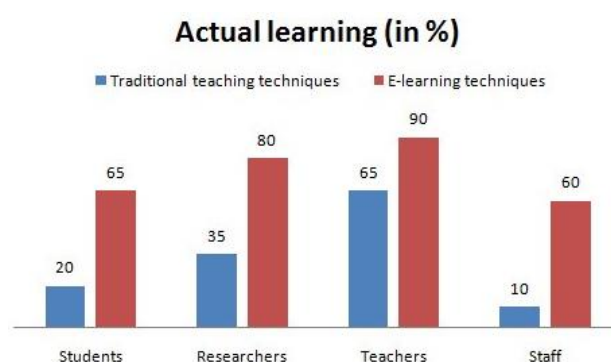


Figure 1- Gaur, S., Chaudhary, A., & Mittal, M. (2015). *A Comparative Study of E-Learning Technique with Traditional Teaching Techniques.*

Concomitant to Virtual Reality, Artificial Intelligence (AI) shows great potential for education, especially when combined. Recent research by Nitin Liladhar Rane (2023) explains that AI can create interactive, accommodating and learner-centric environments. It can adjust content, based on the learner's progress, thus improving engagement and comprehension (Rane, Choudhary, 2023). A convergence of human and AI, promotes the development of both technical and soft human skills such as critical thinking, creativity, emotional intelligence, and complex problem-solving. However, ethical considerations and data privacy must be addressed to ensure the user's digital security and to create an inclusive environment that is accessible to all. Artificial Intelligence has also been proven to be effective in increasing user engagement. Aung Pyae (2023) explored the User Experience and Usability of a metaverse platform for learning. The results show that not only has AI improved the user experience, but it also increased the user's motivation and engagement towards learning (Pyae, 2023).

The objective of this chapter is to evaluate the current literature and indicate areas that require more investigation.

1.2 Limitations of Traditional Teaching Methods

For years, traditional educational methods have depended on approaches such as lectures and textbooks, which have served as the foundation for information transfer. However, these strategies are increasingly seen as having difficulties in engaging the modern student. According to recent studies, conventional education frequently leads to limited engagement, and inadequate information transmission, all of which can inhibit the development of critical thinking and real-world applications of taught concepts (Saira, Nishat Zafar, 2021). One key disadvantage of traditional teaching is that it promotes passive learning, in which students function as passive consumers of knowledge rather than active participants in their educational journey. This passive approach can result in decreased engagement and a lack of deep learning, jeopardising the student's capacity to remember and apply material (Saira, Nishat Zafar, 2021). The one-size-fits-all approach, which is popular in traditional techniques, exacerbates the problem by failing to address individual learning styles and demands, preventing thorough knowledge.

In sharp contrast, current research indicates that technology-assisted learning approaches, such as Virtual Reality, have the potential to overcome these educational barriers. VR offers a highly immersive learning environment in which students may interact with virtual items and see real-time results, enabling an active learning process that traditional techniques frequently lack (Tiffany Leung, Farhana Zulkernine, Haruna Isah, 2018). VR, which aligns with constructivist learning theory, helps learners to actively create information via exploration and experimentation, resulting in increased retention and understanding. Saira (2021) also acknowledges the limits of traditional education in terms of student involvement, an area where as discussed previously, VR thrives. The demanding and realistic quality of VR settings has been demonstrated to boost motivation and engagement, resulting in better learning outcomes (Tiffany Leung, Farhana Zulkernine, Haruna Isah, 2018). This is especially important in the context of transdisciplinary learning, where VR can enable cooperation and experience sharing that traditional techniques cannot (Tiffany Leung, Farhana Zulkernine, Haruna Isah, 2018).

Furthermore, VR exceeds traditional education techniques in terms of safety and accessibility. By removing the limits and hazards associated with traditional learning settings, for example, during lab sessions, VR provides a safe and accessible platform for study, particularly in sectors involving dangerous chemicals or procedures.

1.3 Recent Advances in VR-Assisted Learning

The historical rise and progress of VR platforms have consistently enhanced educational experiences. *Google Expeditions*, which debuted in 2015, represented a watershed event in the integration of VR into education. It enabled immersive field excursions and interactive learning experiences, which have since been enhanced by innovations in 2020 and 2021, such as the integration of AR experiences and improved visual content quality (Informatics, 2023).

Unimersiv and *Alchemy VR* joined the VR educational arena in the early and mid-2010s, respectively, offering interactive experiences on a variety of topics. Both platforms have made significant contributions to the diversity of VR content by making complicated disciplines such as history and biology more accessible through immersive narrative and visualisation.

zSpace released its VR learning platform in the early 2010s, and it has been updated several times, most recently in 2021 and 2022, to include collaborative capabilities and interactive simulations. These improvements have enabled students to explore STEM fields via hands-on experiences in a virtual environment (Informatics, 2023).

They illustrate how virtual reality has transformed the distribution of content information, recreating scientific processes, and visualising abstract concepts, making learning more approachable and relatable to a wide range of people.

These systems show a clear progression of VR's potential in education, from passive material distribution to active, personalised, and immersive learning environments. VR technological improvements have not only made education more engaging and dynamic, but they have also closed accessibility barriers, allowing students from all backgrounds to participate in high-quality learning experiences (Chen, 2023).

1.4 Integration of AI with VR in Educational Settings

The integration of Artificial Intelligence (AI) with Virtual Reality (VR) in educational settings is proving to be a significant advancement in enhancing learning experiences. Highlighted by Zhang and Aslan (2021), this combination is not just a technological upgrade, but a paradigm shift in educational methodologies. AI's ability to personalize learning experiences, coupled with VR's immersive environments, facilitates a deeply engaging and effective education process. Research indicates that AI-enhanced VR applications can dramatically improve student engagement and learning outcomes. For instance, personalized learning environments using AI and VR have shown promising results in computer education, improving both the experience and outcomes for students (Zhang & Aslan, 2021). Studies also suggest that these technologies make learning more intuitive and memorable, particularly in complex subject areas like microbiology, where students can interact with three-dimensional models of microscopic organisms (Rane et al., 2023). However, their study lacks empirical data, relying more on theoretical projections which may not fully represent real-world educational outcomes.

Rane et al.(2023) focus on specific applications within microbiology education, showing how AI-enhanced VR can improve understanding of complex subjects. This study underscores the effectiveness of AI and VR in making intricate content more accessible and engaging. Yet, like Zhang & Aslan(2021) , it does not address long-term impacts on learning retention or the potential cognitive overload from high-stimulus VR environments.

To demonstrate how AI enhances the learning experience in BioVRium, I have detailed the process in a flowchart (see Fig.2). The sequence begins when the user initiates a voice

command by pressing 'A'. This action triggers the system to capture and send the voice data to the AI for analysis. Utilizing natural language processing, the AI evaluates the input and offers contextual guidance or tutorials, thereby enriching the user's interaction with 3D models of microorganisms.

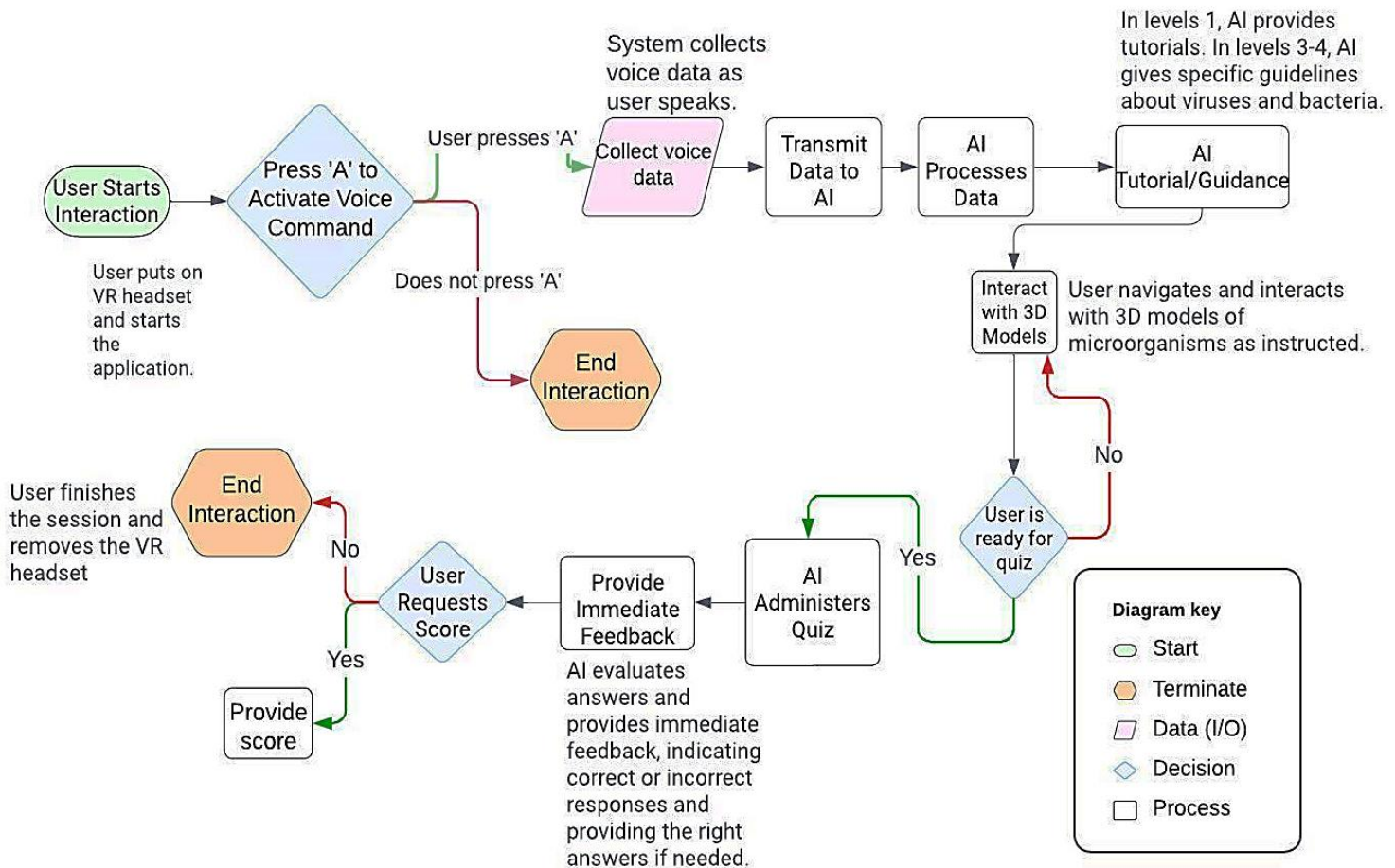


Figure 2 - Flowchart BioVRium AI component.

In more advanced stages, the AI becomes even more interactive. It provides specific instructions and can administer quizzes upon the user's request. Following each quiz, the AI delivers immediate feedback, pointing out the correct answers when mistakes are made. Additionally, upon request, the AI can share the user's quiz scores. This completes the feedback loop, enabling the system to offer customized interactions in future sessions based on the user's performance, thus enhancing the educational experience.

1.5 The Impact of VR on Student Engagement and Performance

During IEEE's conference in 2022 various findings were presented, uncovering that when students are immersed in VR environments tailored with contextual cues aligned to their learning material, their academic performance notably improves (G. Bernal, N. Hidalgo, C. Russomanno and P. Maes, 2022). This form of active learning—characterized by sensory-rich, three-dimensional exploration—echoes the engagement previously discussed in the limitations of traditional teaching methods, effectively addressing the pedagogical shortcomings where traditional approaches may falter.

The study proves the importance of immersing students in VR scenarios that resonate with the content, significantly enhancing their understanding and retention of the material, a concept that is particularly germane to microbiology (Arya, D. Hawes and A., 2022).

Parallel to these insights, Vola (2023) provides empirical support for the positive repercussions of VR on students' academic engagement. Their study within an undergraduate biology lab setting demonstrated that the incorporation of VR not only bolstered students' engagement with complex topics like global biodiversity, but also provided a more immersive experience that students found beneficial for their learning process. Interestingly, their findings suggest that rather than serving as a distraction, VR enhanced students' focus and deepened their connection with the coursework (Vola et al., 2023). Furthermore, the integration of VR led to an increase in grades across various assessments, suggesting that the multisensory, interactive nature of this technology could be a significant factor in elevating students' academic achievements (Vola et al., 2023).

1.6 Review of Related Works: Exploring VR and AI-Enhanced VR Applications in Microbiology Learning

Several applications and studies were found to align closely with this study's topic. Starting with "*Journey to the Centre of the Cell*" (see Fig.3), a pioneering work from 2020, researchers utilized advanced imaging techniques to create a detailed 3D virtual model of a cell. This model enables not just scientists but also students and the public to delve into and interact with an authentic cellular environment. Early testing of this immersive VR environment indicated a notable improvement in the comprehension of cellular functions, suggesting a promising future for VR in education and public engagement.

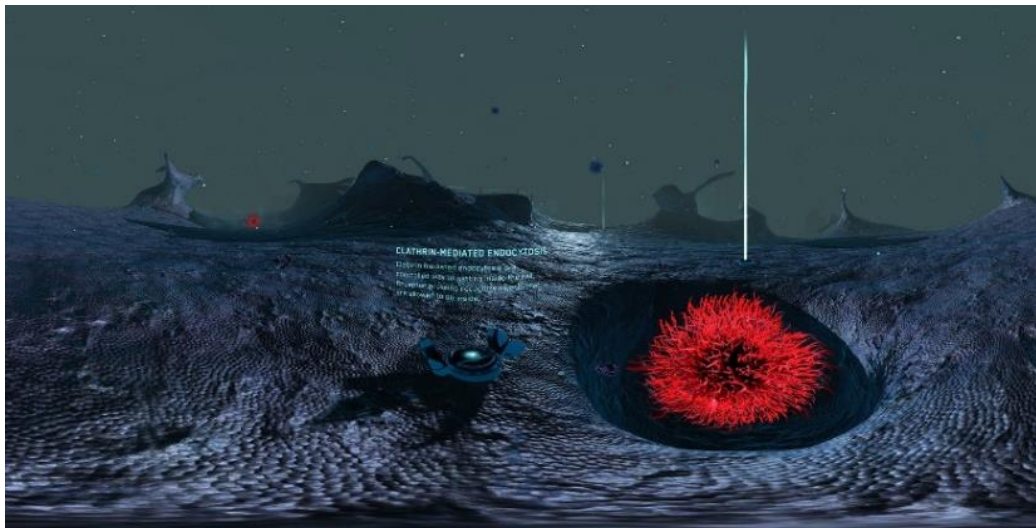


Figure 3 - *Journey to the Centre of the Cell*.

In a similar vein, Slater's "*Transforming Microscopic Data into 3D Models*" (see Fig. 4) addressed the challenges and potential of converting microscopic images into three-dimensional models for educational exploration. The workflow involving 3D Slicer and mesh reduction software (see Fig.5) represents a significant stride in making intricate

microscopic data more accessible and manageable for interactive learning experiences (Slater, Annabel, 2020).

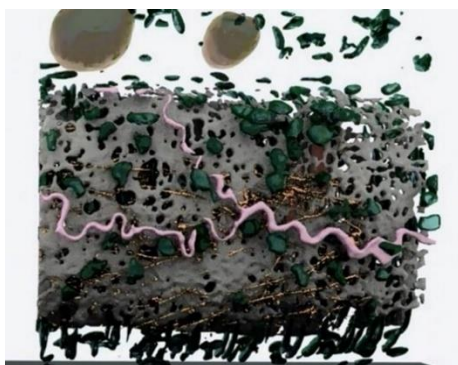


Figure 4 - Slater's 3D Models.



Figure 5 - Microscopic Data into 3D Models.

Although this project has more applications in the scientific sector rather than in the educational one, they have created a virtual gallery to showcase their collection. Their main goal is to create a virtual space where researchers can collaborate by sharing 3D models and educators can benefit from this, by teaching classes using VR within this platform. The platform has been a success since its first days and is frequently by other scientist to build their collections.

Complementing these works is Labster, a company offering an extensive collection of virtual lab simulations aimed at enhancing STEM education (Labster, 2024). Labster's virtual labs are not only immersive and engaging (*see Fig. 6*) but also leverage Artificial Intelligence to tailor and adapt learning experiences to individual student needs. This AI integration facilitates dynamic learning environments where simulations respond in real-time to student inputs, enhancing educational outcomes by providing personalized feedback and support.

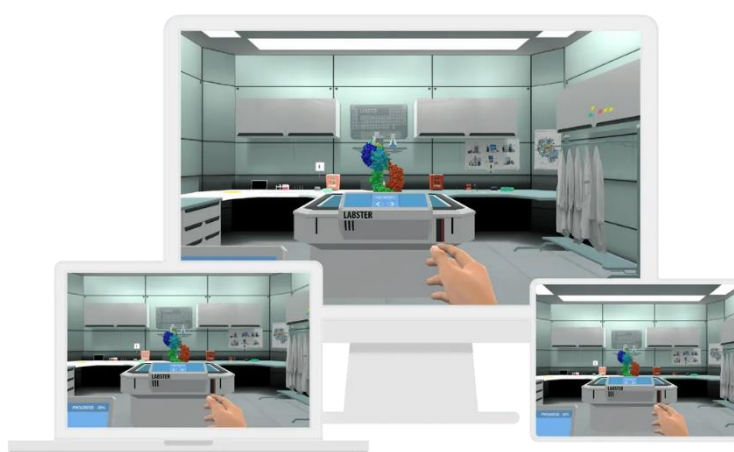


Figure 6 - Labster's User Interface in a laboratory.

Furthermore, Labster addresses real-world challenges by providing students with practical insights into potential STEM careers. The platform's accessibility, aligned with the mission of democratizing education, ensures that complex scientific concepts are

approachable from any location and on various devices (Labster, 2024). The parallels between Labster's offerings and this dissertation's focus on interactive learning within microbiology are evident. Both initiatives utilize VR, enhanced by AI, to create hands-on educational experiences, making STEM learning more engaging and accessible. Labster's virtual labs allow students to grapple with STEM concepts through interactive and adaptive learning processes, echoing this dissertation's aim to enable students to interact with microscopic organisms in a similarly engaging and personalized manner.

1.7 Literature Gap

While existing research establishes a robust foundation for the use of Virtual Reality in educational settings and highlights its potential to transform learning experiences across various disciplines, there remains a significant gap in the integration of Generative Artificial Intelligence with VR. The integration of AI can potentially address specific pedagogical needs by offering personalized learning experiences and adaptive content delivery, yet empirical studies focusing on this integration within the microscopic aspect of biology are notably scarce. Existing studies often concentrate on broader scientific topics or explore the impact of VR on more visually accessible and less complex areas of biology, rather than on microbiology, which is crucial for a deep understanding of cellular processes. Additionally, there is a dearth of comprehensive research on the design, implementation, and evaluative methodologies of VR applications that are enhanced by AI for learning microscopic biology. The literature often lacks detailed discussions on how AI-integrated VR experiences can align with curricular objectives and accommodate diverse learning styles.

2 Motivation and Research Questions

As mentioned earlier, this project aims to develop and evaluate an AI-enhanced Virtual Reality educational platform. This investigation is designed across two versions of the platform: one enhanced with AI characters and one without. This design aims to facilitate a comparative analysis of learning outcomes and find insights related to the interaction between users and the AI characters. The AI-enhanced version offers users the choice among three distinct AI guides—a senior professor (Dr.Ellis), a young knowledgeable woman (Anca), and a cartoon character (BioBear). Each character has a different and unique personality and teaching style to accommodate diverse learning preferences and maximize effectiveness.

The decision to include a variety of AI characters BioVRium is supported by recent research findings, which highlight the significant role of avatar resemblance in building trust during virtual interactions (Tang & Bashir, 2023). Studies have demonstrated that avatars closely mirroring the user's own identity not only create deeper feelings of presence but also enhance trust in the virtual environment. A high level of trust is important when interacting with Artificial Intelligence, especially in educational settings; as explained by Tang & Bashir (2023), the relationship between users and virtual avatars is intricately linked to the dynamics of trust.

As mentioned, a secondary aim is to assess whether the user would trust more and thereby choose one of the 2 human-like characters (Anca and Dr.Ellis) or the cartoonish character (BioBear) which evaluates whether the users would prefer a gamified element within an educational environment. The principles derived from the work of Tang and Bashir (2023) suggest that creating a strong connection between users and avatars can significantly boost feelings of trust and engagement in virtual settings. In the context of using guide avatars, this could mean designing these avatars in a way that is relatable and can effectively guide users in the virtual world, key ingredients that contribute to building trust in virtual settings. This reinforces the importance of designing AI characters that can resonate with users on a personal level, enriching their educational journey in the virtual world.

The core aim is to dissect how the inclusion of generative AI could potentially shift learning paradigms within scientific education by enhancing learning efficacy, usability, and user experience when compared to traditional VR approaches. This exploration is anchored on several objectives:

- Investigating current VR applications within educational settings, particularly in science education.
- Defining the essential design requirements for VR systems that incorporate both 3D modelling of biological organisms and AI guides.
- Evaluating the system's usability and overall learning effectiveness.
- Analyzing user preferences for different AI guide personas along with their impact on learning outcomes and initial user impressions.

The hypothesis of this study is that the inclusion of generative AI in a VR platform leads to higher learning outcomes and usability levels, compared to a traditional VR approach.

Research questions that guided this study include:

RQ1) Does the inclusion of generative AI in a VR platform lead to higher learning outcomes and usability levels compared to a VR-Only approach?

- RQ2)** How do the interactive and AI-enhanced features of the VR platform influence students' engagement and perceived learning effectiveness in studying microbiology?
- RQ3)** What are initial user impressions of engaging with AI characters in a VR educational setting, and which AI character do participants prefer? Why do they prefer that character?

In the subsequent sections, key concepts essential to understanding the integration of Virtual Reality and Artificial Intelligence in educational contexts are introduced. Starting with **Study 1**, where collected user requirements have been analyzed, followed by definitions of 'Virtual Reality' and 'Artificial Intelligence', with a particular emphasis on how these technologies are applied to enhance learning in microbiology. The VR educational platform developed for this study is also described, highlighting its dual configurations: one with AI integration and another without. **Study 2** evaluates the platform's usability by employing a cross-over design that is tailored to address the set research questions.

This study involves two conditions:

- The control condition, where participants engage with the VR platform without AI enhancements.
- The experimental condition, where the same participants interact with the AI-enhanced version of the VR platform.

This design allows for a direct comparison between the two modalities to investigate the previously stated research questions.

3 Study 1 – User Requirements Gathering

3.1 Survey

Through a verified Discord channel for Virtual Reality with researchers, a survey has been distributed for 1 week to gather user requirements for BioVRium.

3.2 Participants

The survey involved 31 participants that heavily influenced the platform's development. The largest group of participants are those aged 25-34 (15 participants), followed by those aged 35-44 (11 participants) (see Fig. 7). Furthermore, most participants (83,9 %) stated they had previous experience with VR systems (see Fig. 8), which increase the significance of these findings.

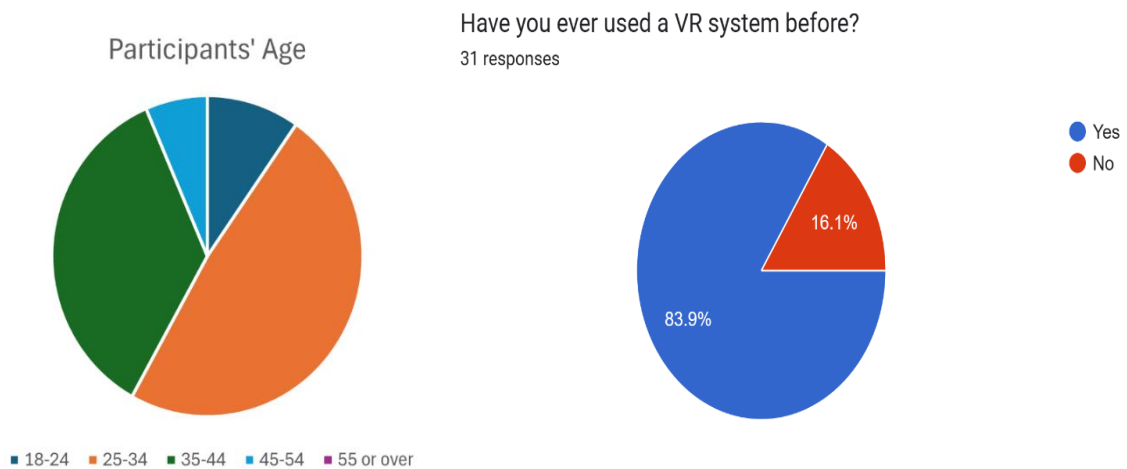


Figure 7 - Study 1, Participant's Age Range.

Figure 8 - Study 1, Participant's Experience with VR.

3.3 Survey Findings

3.3.1 Features

When asked about the most essential features of a VR educational platform for microbiology, the majority - **54.8%** (see Fig. 9) chose 'All of the above,' emphasizing the need for realistic 3D models, interactive content, and adaptive learning paths, which were subsequently incorporated into the platform.

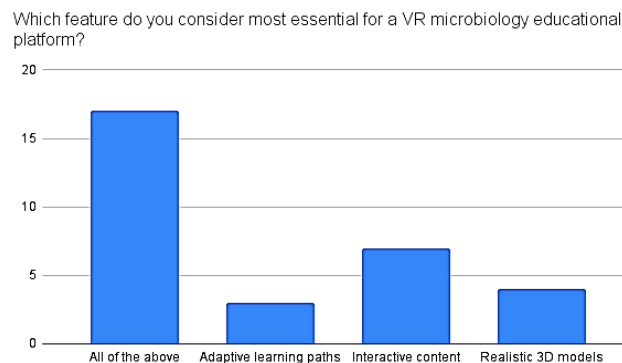


Figure 9 - Study 1, essential features for an educational platform.

Participants clearly highlighted the importance of having built-in quizzes to track their learning progress (see Fig. 10). The vast majority (**32,3%**) rated built-in quizzes and

assessments as 'Very important', leading to their integration in both the AI-enhanced and control versions of the platform.

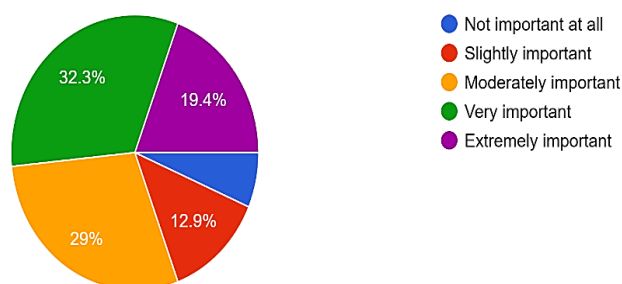


Figure 10 - Study 1, Built-in quizzes.

3.4 Design

Additionally, when queried on the type of VR content beneficial to their microbiology studies, a significant majority - **76.7%** (see Fig. 11) opted for a balance of content that complements and extends beyond their current curriculum, prompting the inclusion of extensive informational resources alongside standard module content.

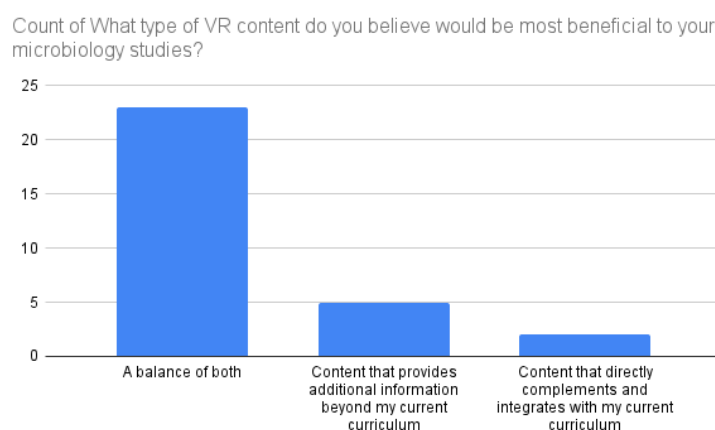


Figure 11 - Study 1, type of content most beneficial for VR platform about learning.

Participants also mentioned about the importance of visualizing and interacting with 3D models:

P6: “Just make sure you can view models from all angles easily by both walking (physically and with a controller stick) and by grabbing (I assume viewing structures is why you'd want to use VR at least)”.

P29 described the importance of feeling immersed, which strengthens the concept of maximizing VR’s immersive component with the aid of strategic design:

“I’ve used VR for a virtual tour of a museum, and it was amazing! If the VR platform can make me feel like I’m inside a cell or a bacteria, that would be awesome”.

3.5 Conclusion

This study has provided significant insights regarding the participant's needs and preferences for BioVRium. The survey, which involved 31 participants highlighted the need for 3D models, interactive content and adaptive learning paths. The emphasis on built-in quizzes for tracking learning progress and the strong desire for 3D models have guided the development of the platform.

4 Software Design and Development

4.1 VR design

4.1.1 Conceptual Foundations

The design of BioVRium integrates high technological specifications with a thoughtfully designed educational framework.

The platform was designed based on constructivist learning theories to promote knowledge-building by engaging learners actively and providing meaningful interactive experiences. Constructivism asserts that learners acquire knowledge most efficiently by engaging with the world and reflecting on their experiences, rather than learning in isolation (Zajda, 2021). BioVRium platform facilitates experiential learning by incorporating interactive 3D models of microorganisms. Users are able to grab the 3D models, rotate them or even grab 2 models at the same time and compare them while reading information about their structure.

In addition, cognitive load theory was instrumental in influencing the platform's design. This idea highlights the significance of effectively controlling the cognitive load throughout learning processes to avoid overwhelming learners and enhance their ability to fully retain knowledge (Sweller, 2011). The platform delivers material in easily understandable parts, hence improving comprehension and retention (see Fig. 12). The information used to create the flashcards is sourced from textbooks and slides used in microbiology modules.

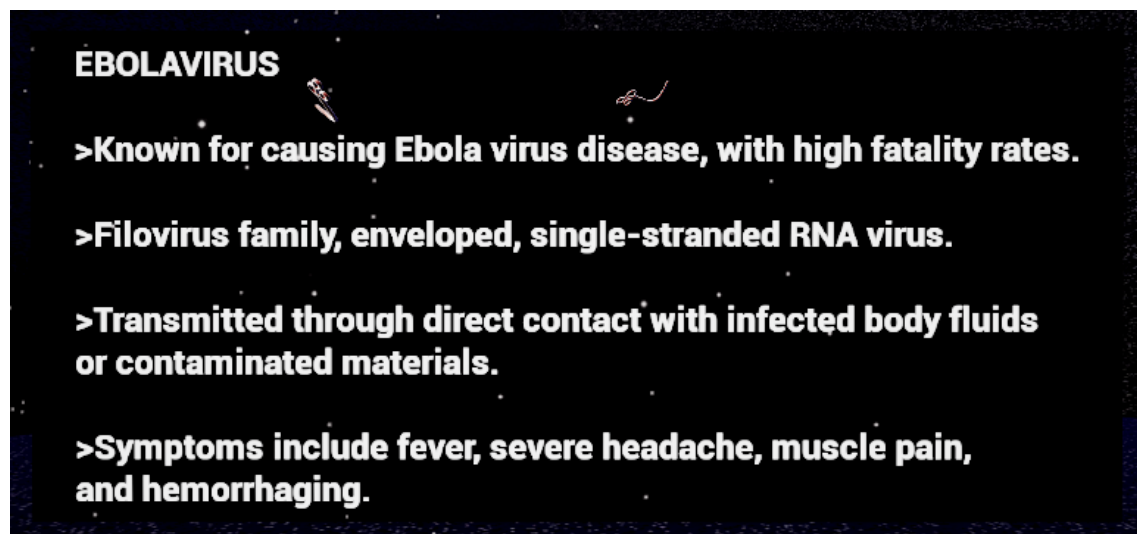


Figure 12 -Information delivered following the cognitive load theory.

4.1.2 Technical Requirements

The platform's development was facilitated by the utilisation of a high-performance computer to effectively manage the requirements of the complex virtual reality content. For this project, it was used a personal computer equipped with an **11th Gen Intel(R) Core(TM) i9-11900KF processor, 32.0 GB of RAM and a 64-bit operating system** on a processor based on the x64 architecture.

For the testing period, the headset used was the Meta Quest 2, which was provided by the project's supervisor.

4.1.3 System Requirements

Educational Efficacy: The platform must deliver microbiology content effectively, including accurate 3D models of microorganisms, content related to the present models and tests to assess learning outcomes.

Interactivity: The platform must allow users to manipulate 3D models, engage with AI tutors, and receive real-time feedback.

Technical Performance: The platform must run smoothly without significant latency or bugs.

User Accessibility: The platform must provide clear instructions, intuitive navigation, and simple controls that accommodate users with varying levels of VR experience.

The survey for user requirements (see Section 3.3, see Appendix B) gathered essential insights regarding the ideal length for a VR session. From the average user's perspective, it should be approximately 30 minutes (see Fig. 13). In the development BioVRium, the decision regarding the session's duration was informed by both current literature and user feedback. Sumardani and Lin's (2023) study suggested that the optimal duration for virtual reality learning is less than 15 minutes. However, the participants indicated a preference for longer sessions, with an average suggestion of 30 minutes. Taking both these factors into account, the sessions last 20-25 minutes. This duration aims to provide an effective learning experience while also considering user comfort and preventing cognitive overload.

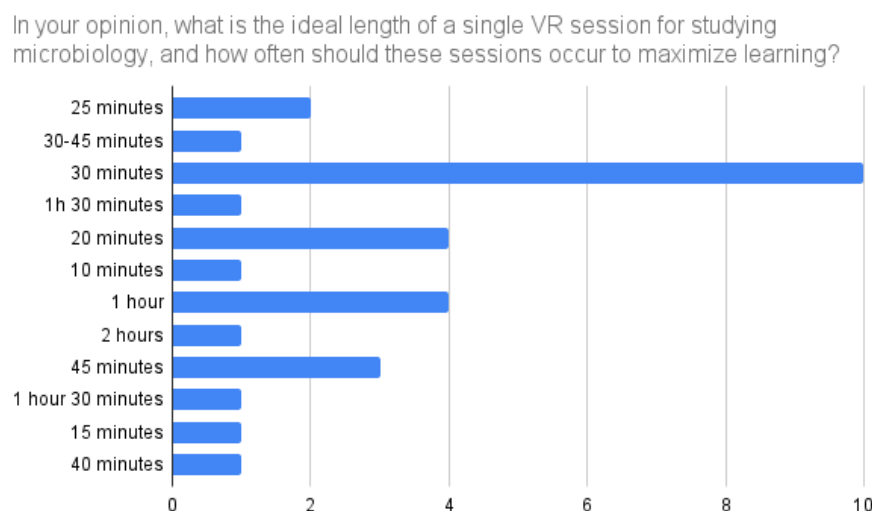


Figure 13 - User Requirements Gathering.

4.2 Design and Technical Structure

4.2.1 BioVRium Architecture

The educational content within the platform is specifically designed to be beginner-friendly, ensuring that even participants without prior knowledge can grasp the fundamental concepts of microbiology and it is organized into four progressive levels:

Level 1 serves as an introductory setting within a virtual science classroom, helping users familiarize themselves with the VR environment and interface (see Fig. 16). A notable

feature in this level is the microscope, which, upon interaction, teleports users to the subsequent level.



Figure 14 - Level 1 – science classroom.

Level 2 acts as a gateway, situating users inside the microscope amidst red blood cells, with portals leading to distinct realms dedicated to viruses and bacteria (see Fig. 17).

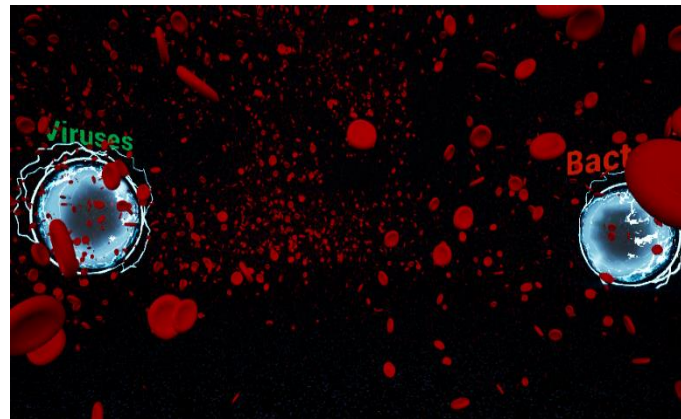


Figure 15 - Level 2 – inside the microscope.

In *Level 3*, users exploring the virus realm encounter 3D models displayed in a museum-like U-shaped layout, encouraging interactive learning about each virus (see Fig. 18).

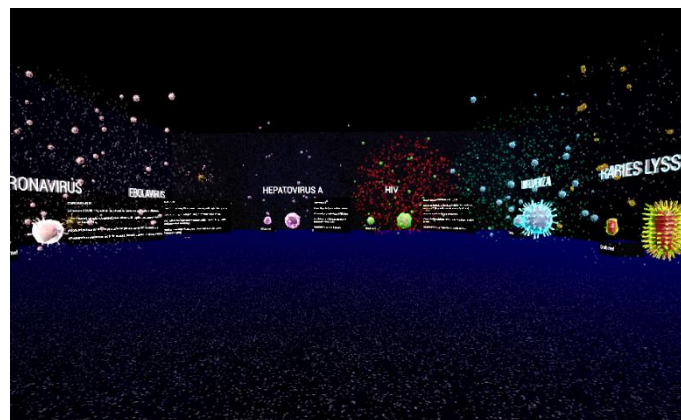


Figure 16 - Level 3, inside the Virus world.

Similarly, *Level 4* guides users through a bacterial exhibit, where they can interact with and learn about different bacterial models arranged in a corresponding U-shaped layout (see Fig 19).

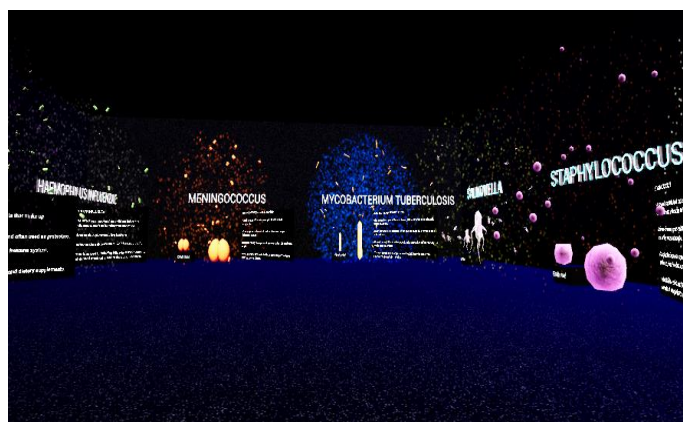


Figure 17 - Level 4, inside the Bacteria world.

4.2.2 Environment

Unreal Engine 5.3.2 was used to develop BioVRium. Central to the platform are 16 highly detailed 3D models of viruses and bacteria (see Fig. 14, see Fig. 15).

The models were sourced from a comprehensive package on the Unreal Engine Marketplace called "Microscopic Organisms." This package included not only the models but also the materials and animations associated with each organism. The animations were crucial to creating realistic representations of the microorganisms' movements.

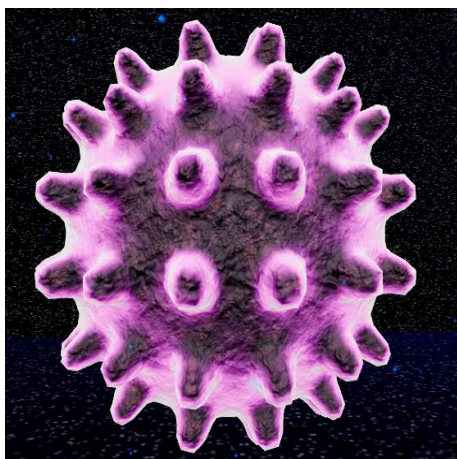


Figure 18 - Hepatovirus A.

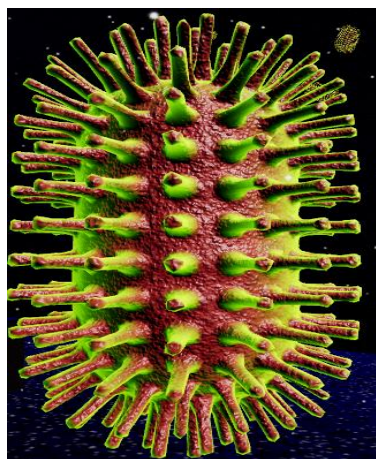


Figure 19 - Rabies Lyssavirus.

Highly realistic models have been chosen for this project because they are the key in providing accurate representations, allowing students to observe and interact with models that exhibit realistic biological behaviours.

The first level, where users begin their journey, is developed using the "Chemistry Class - School" package, also acquired from the Unreal Engine Marketplace. The portals used to travel from one level to the other are also part of a package acquired from the marketplace, "Portals VFX Pack."

The integration of these marketplace assets was strategically managed to ensure they seamlessly blend into BioVRium's educational aims. Each asset was carefully evaluated and tailored to fit within the platform's design, ensuring consistency in visual style and educational purpose. This process involved adjusting the positioning, scaling, materials behaviour and some of the animation parameters of the 3D models to optimize them.

Interactive elements within the platform are designed to facilitate an engaging and hands-on educational experience. The platform leverages the interactive potential of VR by enabling students to approach, inspect, and conceptually manipulate the 3D models. This direct interaction is supported by intuitive VR controls and guided exploration facilitated by the platform's AI-driven characters.

4.3 Synergizing VR and Generative AI to Enhance Microbiology Learning

In this study, the integration of Virtual Reality and generative Artificial Intelligence is central. As mentioned in the previous chapter, VR is utilized to create a fully immersive three-dimensional environment where students can interact with detailed models of microscopic organisms.

Artificial intelligence enhances the VR experience by introducing a guide that mimics a real-life teacher. It integrates an interactive approach to learning new information directly from the AI characters, offering tailored learning styles provided by the available AI guides and providing immediate feedback during a quiz. Furthermore, the AI algorithms analyse student interactions within the VR environment to tailor the educational content dynamically. For instance, if a student struggles with understanding certain microbial functions, the AI can modify the presentation of the content to simplify the information or even suggest methods to reinforce retention. Additionally, the AI characters can respond to student inquiries in real-time regarding anything, even if it is beyond the content offered in the platform. They can offer supplementary explanations, guide interactions with the 3D models, or provide feedback based on the user's progress.

The choice of combining AI and VR for this application is driven by its potential to make learning more engaging, intuitive, and customized (Bermejo, 2023). The immersive nature of VR helps bridge the gap between theoretical knowledge and practical understanding, while AI's adaptability ensures that the educational journey is aligned with individual learning needs and speeds (Tiffany Leung, Farhana Zulkernine, Haruna Isah, 2018). This dual-technology approach aims not only to enhance academic performance in microbiology but also to ignite student curiosity and motivation.

4.4 AI Characters in BioVRium

4.4.1 Character Description

BioBear, an animated teddy bear, is designed to transform learning into a playful and engaging experience (see Fig.20). His presence in the VR environment is intended to reduce the intimidation often associated with complex scientific subjects like microbiology. BioBear's approachable and friendly nature makes it particularly appealing to users who might find traditional educational settings daunting or who respond well to gamified learning environments.



Figure 20 - BioBear.



Figure 21 - Anca.

Anca is designed as a young, knowledgeable microbiologist who combines expertise with a relatable and friendly demeanour (see Fig. 21). Her character is intended to bridge the gap between rigorous academic knowledge and modern, interactive learning methods. Anca is particularly suited to learners who seek a balance between credible scientific figure and a supportive, engaging learning experience.

Dr. Ellis represents the archetype of a traditional academic expert (see Fig. 22). Portrayed as a senior, knowledgeable professor Dr. Ellis appeals to learners who prefer a more structured and authoritative approach to education. His persona is crafted to reflect deep expertise and reliability, aiming to attract users who appreciate thorough explanations and a serious educational tone.



Figure 22 - Dr. Ellis.

4.4.2 Rationale for Character Diversity

The rationale behind these distinctly different characters was to not only cater to diverse user preferences but also to investigate how different demographic representations in AI characters—such as gender and age—impact user choices and interactions. The inclusion of both an older male character and a younger female character, alongside a non-human cartoon character, allows the study to explore whether user preferences are influenced by stereotypes or relatability to the characters' personas. This diverse setup provides valuable insights into how identity and presentation in educational AI can affect learning engagement and effectiveness, offering a deeper understanding of user dynamics in virtual learning environments (D. Hepperle, 2021).

4.5 Interactions with Generative AI in BioVRium

Upon entering level 1, users find themselves in a virtual science classroom, greeted by the AI character that they chose. A visual cue on the right wall guides users to initiate interaction by pressing the "A" button and request for a tutorial. The guide, whether it's BioBear, Dr. Ellis, or Anca, introduces itself and provides essential safety instructions, navigation guidance, and an overview of its role as an educational guide. This initial interaction not only acquaints users with the VR mechanics but also sets up a continuous interactive framework for the subsequent educational content. As users advance to more complex levels focusing on viruses and bacteria, the guide's role broadens: it includes specific guidance on how to interact with the 3D models and gives instruction to the user on how to navigate the module. Users can manipulate these models, examining them from various perspectives while the guide provides contextual scientific insights, thereby enhancing the learning experience through guided exploration. Moreover, the guide assesses user readiness for evaluations, transitioning users to quiz modules upon their verbal indication of readiness, and further supports learning by offering immediate feedback on quiz responses (see Fig.23). Throughout the educational sessions, the guide remains available to answer any inquiries, ensuring that students can obtain real-time clarifications and extend their learning beyond the structured content.



Figure 23 - AI Character in Level 3.

4.6 Exploring CONVAI: The AI Behind BioVRium

CONVAI is a plugin available to use with Unreal Engine, however it is much more than that: it is a sophisticated system that is not merely a scripted set of responses, but a dynamic entity equipped to handle and adapt to a wide range of user interactions.

Operating on the cutting edge of technology, CONVAI incorporates advanced algorithms that allow it to process user inputs in real-time and respond in a manner that simulates natural human interaction. This capability is rooted in techniques that enable CONVAI to understand and interpret the spoken words of users, discerning their queries, commands, and even their nuances of language (CONVAI, 2024). This understanding is crucial for the guide to serve not only as a guide but as an interactive tutor capable of addressing specific educational needs and questions posed by the users (CONVAI, 2024). The guide's response mechanism is designed to be contextually aware, meaning that CONVAI can assess the current learning scenario within the VR environment and provide feedback or information that is most relevant to the user's immediate experiences (CONVAI, 2024).

Moreover, CONVAI is designed to adapt its instructional strategies based on the user's performance and progress. Utilizing embedded analytics, the system can evaluate user responses to quizzes and interactions, adjusting the complexity and depth of subsequent content to suit their learning pace (CONVAI, 2024). This adaptive learning approach ensures that the educational content is not only challenging enough to keep the user engaged but also aligned with their capacity to absorb and understand complex scientific concepts (Neil Sardesai, 2024). In addition to its educational functionalities, CONVAI also enhances BioVRium by managing the user interface (UI) within BioVRium. It assists users in navigating the VR space, providing guidance and ensuring that the VR experience is smooth and intuitive.

4.6.1 Architecture

CONVAI is built on a modular architecture that allows for scalability and flexibility in its operations. The core of CONVAI's functionality is powered by a combination of natural language processing (NLP) and machine learning algorithms, which are implemented using Python and integrated with TensorFlow, a popular open-source machine learning library.

- **Natural Language Processing (NLP):** This system utilizes models trained on extensive datasets of terminology and common language to accurately interpret queries and commands. Techniques such as tokenization, parsing, and semantic analysis are employed to decompose and understand the user's spoken or typed inputs (Chowdhary, 2020).
- **Machine Learning for Adaptive Learning:** CONVAI uses supervised learning algorithms to adapt educational content dynamically. By analysing user responses and interaction patterns, the system adjusts the complexity and delivery of instructional content. This adaptive approach is facilitated by a decision-tree algorithm that categorizes user proficiency levels and tailors interactions accordingly (Konstantinos Pliakos, Konstantinos Pliakos, Jung Yeon Park, 2019).

CONVAI's main challenges that were also present during the platform's development are latency and accuracy. It has been observed that at times, the AI NPC takes longer than expected to respond, even if the wait time is just a few seconds. As a result, it can lead to a decline in user experience. In addition to this, non-native English speakers may encounter inaccuracies in the responses that the AI NPC provides.

4.6.2 Process

To develop the three AI characters—Anca, Dr. Ellis, and BioBear, a structured approach was undertaken, beginning with the configuration of the avatars. Following the initial setup, each character was created with a unique background story and a distinct personality. After character development, comprehensive documentation was crafted to support each AI's interactions within the platform. This included the creation of four key documents: the Game Structure, Tutorial, Virus Quiz, and Bacteria Quiz, which were integrated into each character's knowledge bank. This integration was crucial in ensuring that the AI could retrieve and utilize relevant information during interactions with users. Rigorous testing phases were conducted to refine the AI responses, ensuring accuracy and appropriateness. These testing sessions were critical in perfecting the characters' ability to deliver educational content effectively and interact dynamically with users.

In terms of user interaction with the AI, the conventional method involved pressing the "T" key on a computer's keyboard. However, for the BioVRium platform, tailored to a VR environment, this approach was suboptimal. Consequently, a specific data asset (IA_TALK) was developed within the Unreal Engine framework to facilitate a seamless interaction mechanism. This development linked the action of talking to the AI with pressing the "A" button on the right-hand controller of the VR headset. Modifications to the underlying code were necessary to implement this change, enhancing the user experience by aligning interaction methods with the intuitive controls of VR hardware.

5 Study 2 - Usability Evaluation

5.1.1 Operational Set-up

Due to the university not being equipped with the necessary VR hardware or computers with the required specifications, I provided the equipment needed for the study. On each day of testing, I brought my PC, monitor, keyboard, mouse, VR headset and microphone (see Fig. 24) to ensure the platform could operate smoothly without technical difficulties. The setup was installed in one of the University's available study rooms (PE.02.009 Library Study Room One).



Figure 24 - Testing Set-up

Each session with the participants lasted approximately 60 minutes. Before starting, participants underwent a brief orientation to familiarize themselves with the VR equipment and the study procedures. This included instructions on how to safely use the VR headset and navigate the virtual environment.

5.2 System Validation

Functional testing primarily focused on ensuring that the VR application operated correctly. This included:

Launch and Operational Checks: Verifying that the application launched successfully on the PC and on the Meta Quest 2 and functioned without major errors. This check confirmed base functionalities, such as scene loading and basic user interactions with the VR environment.

Interaction Testing: Manual testing was performed to confirm that interactions with in-game elements, such as 3D models and navigation within the VR space, were responsive and functioned as designed.

Technical Review of AI Implementations: The AI component, CONVAI, underwent a series of internal tests to validate its performance. These tests involved:

1. *Scripted Interaction Simulations*: Using predefined scripts to simulate user interactions with the AI system to ensure it responds accurately and appropriately.
2. *Logic and Flow Verification*: Reviewing the logical flow within AI conversations to ensure that the AI can sustain coherent and contextual dialogues.

The study employed a two-week crossover design, between two groups: control (VR without AI) and experimental (VR enhanced with AI) that tested both conditions. This design allowed each participant to interact with both conditions and each condition-topic combination is presented first and second an equal number of times, which helps to control for potential confounding variables, such as individual differences in learning style or prior knowledge. The crossover framework was chosen to control for order effects and to ensure each subject experienced both modalities of the educational tool—control and experimental.

5.3 Participants

Recruitment of participants for the study was conducted using a targeted approach within the University of West London. To effectively reach potential volunteers, a flyer containing key information about the study was distributed across various locations within the university premises (see Appendix A). The flyer featured a QR code that, when scanned, directed interested individuals to a Google Form. This form was designed to gather essential information from the participants, including their full name, email address, experience with Virtual Reality (VR), field of study, and their availability to choose from predetermined dates and times for the testing sessions.

From this recruitment effort, 18 individuals initially signed up to participate. However, only 10 individuals completed the entire testing process. To ensure effective communication and confirm participant commitment, all registered individuals received a series of preparatory emails. These emails included confirmation of their booked session, reminders of the study's purpose and procedures, and a pre-assessment survey that needed to be completed before their session.

The pre-assessment was crucial not only for gathering baseline data but also for confirming the participant's commitment to the study. Participants were required to respond to the email and complete the pre-assessment survey to retain their slot in the study. Non-responsive individuals were automatically removed from the roster to maintain the integrity and scheduling of the crossover study design. The final group of participants who actively took part in the testing comprised an equal gender distribution: 5 males and 5 females. This balanced demographic was beneficial in minimizing gender-based variability in the study's findings. Communication with the participants remained ongoing via email throughout the testing period, ensuring they were well-informed and prepared for their participation in the VR sessions.

5.3.1 Age Distribution

The study included participants with a wide age range. The youngest participant was 19 years old, while the oldest was 47 years old. On average, the age of the participants was 28 years. The ages varied around this average with a standard deviation of 8.11 years. This indicates a moderate spread in the ages of the participants in the study (see Table 1).

Statistic	Value
Minimum Age	19 years
Maximum Age	47 years
Mean Age	28 years
Standard Deviation (SD)	8.11 years

Table 1 – Age distribution.

5.3.2 Microbiology Background

A significant majority, 9 out of 10 participants, identified themselves as novices in the field of microbiology, indicating a general lack of in-depth knowledge about this topic before they engaged with the VR experience (Table 2).

Further analysis of the participants' fields of study revealed considerable diversity, with academic disciplines ranging from Pharmacology and Biomedical Science to fields less directly connected to microbiology, such as Computer Science, Creative Computing, Criminology, MBA, and Cyber Security (see Table 2).

Category	Number of Participants
Total Participants	10
Self-Identified Novices in Microbiology	9
Self-Identified Intermediates in Microbiology	1
Fields of Study Related to Microbiology	3
Fields of Study Not Related to Microbiology	7

Table 2 - Participant's Background.

5.4 Study Results

5.4.1 Engagement

By analysing session logs and interaction data, it was assessed whether the inclusion of AI elements influenced participants' willingness and consistency in engaging with the VR content. Furthermore, it was possible to track specific behaviours such as the *frequency of attempts* participants made to interact successfully with 3D models, the *time taken* to complete designated tasks, and their *overall navigational efficiency* in the VR space.

5.4.2 Learning Outcomes

Pre- and post-session assessments were utilized to measure *knowledge acquisition* directly which is a valid tool for assessment (Huang et al., 2020; Kourtesis et al., 2021). The quantitative data allowed for a comparative analysis to determine if participants learned more effectively with the AI-enhanced VR system. Additionally, interaction logs from the AI system offered deeper insights into the areas where participants sought further information, which helps in identifying aspects of the platform that could be most beneficial or would need enhancement.

5.4.3 AI Character Preferences

Upon completing their sessions, participants provided feedback through exit questionnaires that included both structured and open-ended questions. This feedback was important to understand participants' reasons for choosing a particular AI character. The qualitative responses were analyzed to extract information related to the *perceived value of AI* in helping understand complex concepts, the enjoyment of the learning process and the rationale of their choice.

5.4.4 Usability and User Experience

User experience and usability were evaluated through a combination of the System Usability Scale (SUS) and detailed qualitative feedback gathered via the UX form and observational data during the sessions which are valid tools for assessment (Kourtesis et al., 2021; Li et al., 2021). This part of the study specifically examined whether AI integration positively impacts the usability of the VR system and motivates learners more than when using VR alone.

5.5 Data Analysis

The approach was designed to integrate quantitative and qualitative data to provide a comprehensive insight into participants' interactions with the platform. Quantitative data was collected through pre- and post-session assessments to measure knowledge gains and through CONVAI logs that tracked interaction patterns and adherence. Usability was evaluated using the *System Usability Scale* (SUS) and Likert scale questions included in the post-assessment questionnaire. These quantitative data were analyzed using Microsoft Excel. This helped to quantify the learning improvements of each condition and to statistically validate the differences in usability and engagement metrics between the VR-only and AI-enhanced setups.

Qualitative insights were gathered to deepen the understanding of the user experience and to contextualize the quantitative findings. During the testing sessions, observational notes were taken, and all sessions were audio-recorded with the participant's consent.

The analysis of qualitative data was performed using thematic analysis, more specifically using Infranodus. The data were categorized into themes that emerged directly from participants' feedback and were pertinent to key research interests such as: *Interactive and Learning, Complex and Feedback, User-Friendly, Engaging and Interface* (see Fig. 25, see Fig. 26).

These themes were informed by the initial hypotheses about the potential benefits of generative AI integration in VR education and were fine-tuned based on the specific comments and feedback from participants. For example, **80%** of the participants expressed they would use more often the AI-enhanced version rather than the VR-Only version. To provide a holistic view of the study results, the quantitative and qualitative findings were integrated in the final analysis phase. This approach allowed for a richer interpretation of the data, where qualitative insights could explain trends observed in the quantitative analysis or highlight areas where the numerical data might not fully capture participant experiences. Direct quotes from participants were selectively included to illustrate key points. Quotes were anonymized and identified using participant codes (e.g., P1, P2, P3...) to maintain confidentiality while providing transparency.

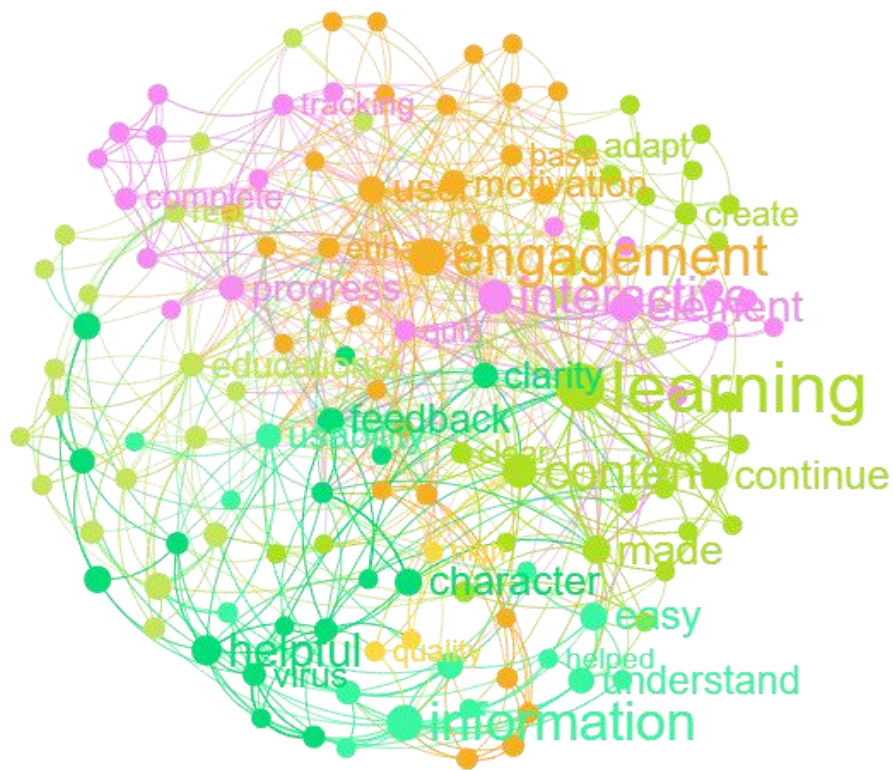


Figure 25- Thematic Analysis.

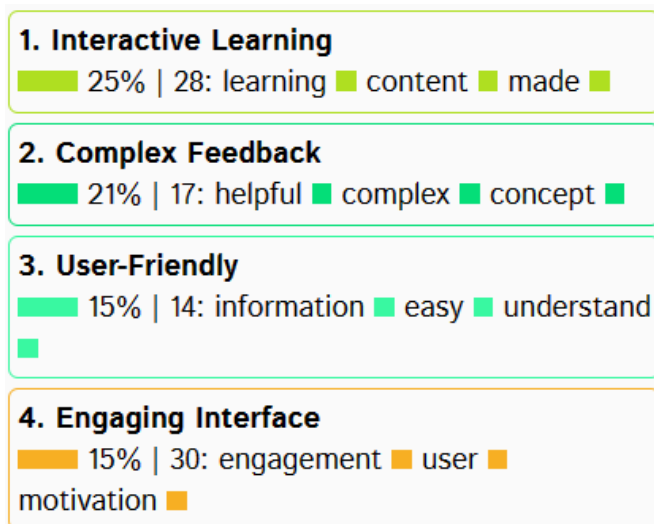


Figure 26- The main 4 themes from the thematic analysis.

6 Result and Discussions

6.1 Engagement

6.1.1 Quantitative Results on Engagement

In this research, Engagement was evaluated through multiple quantitative metrics: *Time Spent in Sessions*, *Interaction Frequency with 3D Models* *Completion of Levels*.

6.1.1.1 Time Spent in Sessions

The amount of time each participant spent while engaging in both versions of BioVRium is a direct indicator of their interest and involvement. Data collected revealed that control group's participants spent an average of 19 minutes out of the allocated 25 minutes (see fig. 27). Some participants either rushed through the tasks due to a lack of interest or had a higher level of knowledge. However, while engaging with the AI-enhanced version, 6 out of 10 participants utilized the full 25 minutes provided. This indicates that the AI-enhanced version was more effective in maintaining participant engagement throughout the entire session. This is mostly due to the participants' desire to interact with the AI guide.

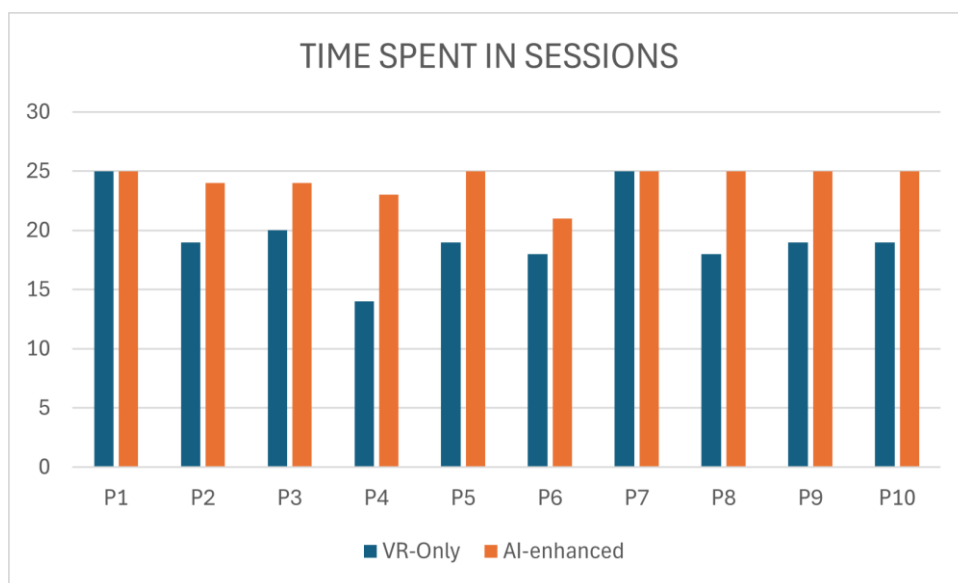


Figure 27 - Time Spent during testing.

6.1.1.2 Interaction Frequency with 3D Models

Interestingly, data showed participants interacted a higher number of times while testing the control (VR-only) version (see Fig. 28). This could be attributed to the absence of the AI guide, which might capture some attention away from direct interaction with 3D models. However, this could also be due to time limitation set for the testing session. It could be argued that, if given more time, the participants might have interacted an equal, or higher, number of times with the 3D models of microorganisms.

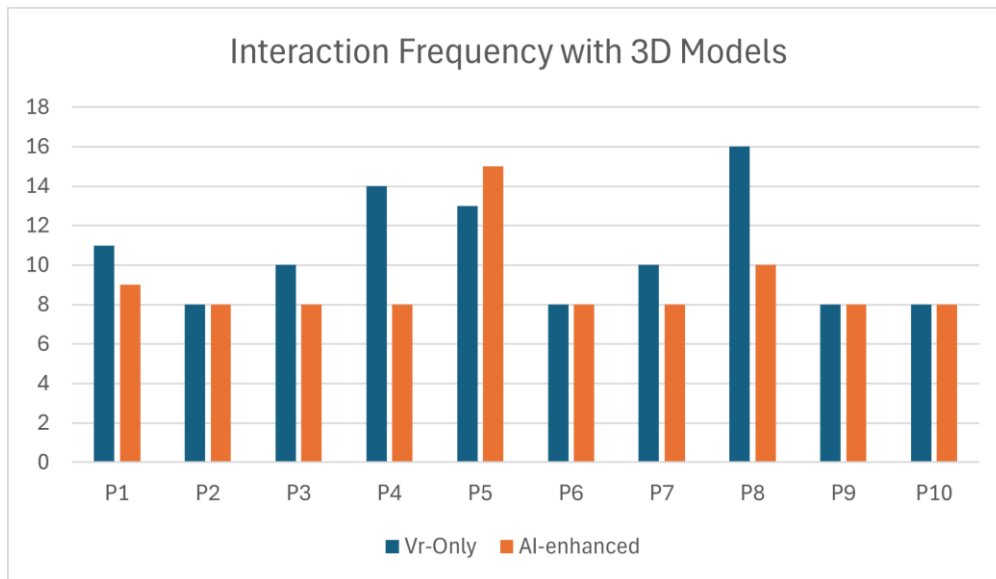


Figure 28 - Bar Chart of Interaction Frequency with 3D Models.

6.1.1.3 Completion of Levels

All participants successfully explored and completed the designated levels in both the VR-only and AI-enhanced sessions. This uniform completion rate across both session types suggests that the basic navigational and operational aspects of the VR platform were well-constructed and user-friendly, allowing participants to achieve the set learning milestones without significant difficulty. This metric, though not differing between conditions, reinforces the platform's effectiveness in facilitating structured educational experiences.

In this chapter, Section 6.1 evaluates the impact of AI on engagement levels within VR sessions, showing that participants spent an average of 6 minutes more when interacting with the AI-enhanced version of BioVRium, **directly addressing RQ2**.

6.2 Learning Outcomes

6.2.1 Pre-Assessment and Post-Assessment Comparison

In the pre-assessment survey, 9 out of 10 participants identified themselves as *novices* in microbiology, while 1 participant described themselves as *intermediate* (see Fig. 29). In contrast, the results from the post-assessment survey highlighted a significant shift in self-reported knowledge levels; 2 participants still considered themselves *novices*, 7 considered themselves *intermediate* and 1 participant described themselves as *advanced* (see Fig. 30). These results mark an improvement in perceived knowledge about microbiology after using BioVRium.

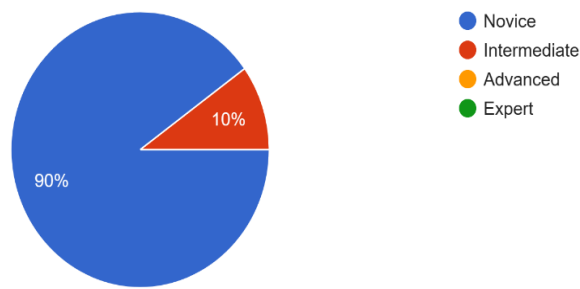


Figure 29 - Pre-Assessment question "What is your current level of knowledge about microbiology?".

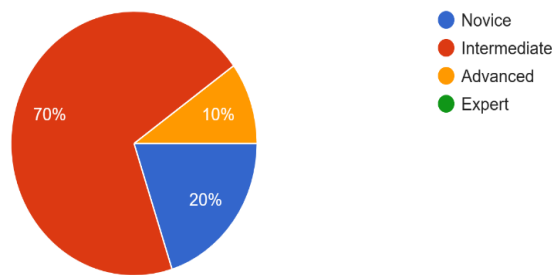


Figure 30 - Post-Assessment question "After using the VR platform, how would you rate your knowledge about microbiology?".

6.2.2 Quantitative Results on Learning Outcomes

6.2.2.1 Mann-Whitney U Test of Knowledge

To test whether there was a difference between the two versions from a learning point of view, each participant was tested after the completion of both standard BioVRium and the version with the AI component. The tests contained 8 questions and they were identical across the two conditions. As mentioned previously, each condition has been tested 5 times with each topic (*virus*, *bacteria*), and each condition-topic combination is presented first and second an equal number of times.

Participants performed slightly better when engaging with the AI-enhanced version and learning guided by an AI character (see Table 3, see Fig. 31).

Participant ID	Vr-Only scores	AI-Enhanced Scores	Difference	AI Guide
P1	6	7	1	BioBear
P2	7	8	1	Dr.Ellis
P3	8	8	0	Anca
P4	7	7	0	Anca
P5	7	8	1	Anca
P6	7	8	1	Anca
P7	8	8	0	BioBear
P8	5	8	3	BioBear
P9	6	8	2	Anca
P10	7	6	-1	Anca

Table 3 - Table 3 - Participant's test score.

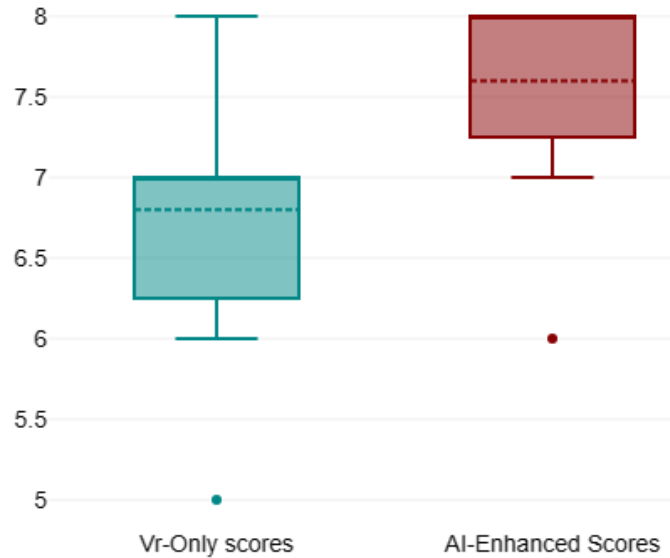


Figure 31 - Distribution of Knowledge Test Scores.

6.2.2.2 Effect size

A **Mann-Whitney U-Test** was conducted to compare scores between Vr-Only scores and AI-Enhanced scores. For the given data, the test showed that the difference between Vr-Only scores and AI-Enhanced Scores, was statistically significant, $U = 24$, $n1 = 10$, $n2 = 10$ $p = .026$ (see Table 4). Thus, the **null hypothesis was rejected**. The mean score for the Vr-Only group is 6.8, meanwhile, the mean score for the AI-Enhanced group is 7.6 (see Fig, 32). This indicates that the values in the AI-Enhanced scores group tend to be higher than those in the Vr-Only scores group.

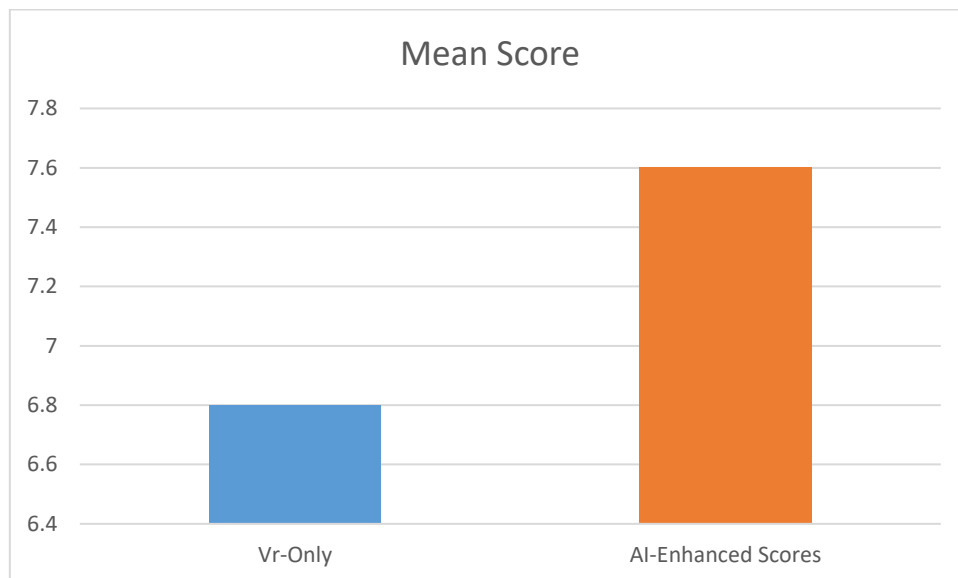


Figure 32 - Mean Score of control group and AI

The effect size r was **0.47**, which is a medium effect. Effect size r according to Cohen (1988):

- $|r| < 0.1$ no effect / very small effect
- $|r| = 0.1$ small effect

- $|r| = 0.3$ medium effect
- $|r| = 0.5$ large effect

Mann-Whitney U-Test – Learning outcomes				
U	z	asymptotic p	exact p	r
24	-2.1	.017	.026	0.47

Table 4 - Mann-Whitney U-Test Results.

These findings not only address directly **RQ1**, but also confirms the hypothesis of this study: the integration of generative AI in traditional VR leads to higher learning outcomes.

6.2.3 Qualitative Results on Learning Outcomes

When asked which version of the software they found more effective for learning about microbiology, participants expressed a range of sentiments about their experiences. For instance, **P1** found the AI-enhanced sessions enjoyable, especially the interaction with the AI guide, Biobear, which brought fun to their study session:

“I enjoyed learning with AI as it brought fun to my study while interacting with the AI”.

P8 had a similar experience, and appreciated the ease with which the Ai, Anca, explained complex topics such as Salmonella:

“I found the version with AI to be more effective. I asked Anca to give me a couple of facts about Salmonella and everything was explained easily. I enjoyed interacting with AI, it was really interesting”.

At the same time, **P9** and **P10** were impressed by the helpfulness of the AI guide in navigating the VR environment.

P9: *“I haven't noticed big differences, but it was a short interaction. Probably, using it for a longer period, the version with AI would be more effective for learning. This was my first time using VR, so asking Anca for a tutorial and how to move around was helpful”*

P10: *“The version with AI. I wish I had more time to interact with the AI help. I definitely believe the version with AI is more effective for learning, you can learn a lot”.*

On the other hand they both expressed that the time-limit was detrimental in their experience with BioVRium. Conversely, some participants faced challenges learning while interacting with the AI component. **P3** mentioned difficulties in needing to return to a specific location to interact with the AI:

“Without AI, interacting with the AI was a bit of a challenge having to go back to where it is to communicate”.

P5 and **P6** commented on the AI's response time. Specifically, they were unsatisfied with the delays in receiving information. **P5**, however, mentioned that overall they prefer the version with AI as being more effective when it comes to learning:

“The one with AI was more effective, because I could always ask for clarification or further help in real time. However, the AI character was very laggy, does not reply instantly and sometimes does not understand what I am saying;. Overall, I think this application if it was fully developed it could help grow my knowledge of microbiology, as

of right now my knowledge remains the same level because it was a relatively short experience and even if I feel I learned a couple of interesting facts, I would not consider myself anything more than a novice.”

P6: *“without AI because I did not need to wait for a long period to know what the answer to the question asked is”.*

Despite these challenges, even critical feedback often acknowledged the potential benefits of AI, suggesting that a fully developed version of BioVRium could even further enhance the learning experience.

6.2.4 Correlation between Quantitative and Qualitative Results

Participants who reported positive interactions with the AI components, such as P8 and P10, generally showed improvements in both and high engagement in their post-test scores. This highlights that if interacted with correctly, AI helps learners achieve better learning outcomes. On the other hand, those who experienced difficulties and frustration with the AI’s responsiveness and accuracy, also showed less improvement during tests, as seen with **P5** and **P6**. This indicates that technical issues might overshadow the full educational potential of BioVRium with AI.

A clear conclusion that can be drawn from these findings is while AI can improve the learning experience, it heavily depends on the quality of interaction with the AI component and the system’s technical performance.

Section 6.2 analyses and compares the learning outcomes for both versions, with and without AI, responding directly to **RQ1**. Furthermore, section 6.1 also answers **RQ2** by exploring how AI-enhanced VR might have influenced students’ perceptions of their learning gains.

6.3 Comparative analysis of AI character preferences

As explained in **Chapter 4**, BioVRium offers to its users the possibility of choosing an AI guide: Dr.Ellis, BioBear or Anca. Out of 10 participants, the vast majority (6) preferred Anca (see Table 5). For a deeper understanding of the character’s design and the rationale behind their creations, readers are referred to **Chapter 4**.

Overview of Preferences	
Anca	Chosen by 3 participants
BioBear	Chosen by 6 participants
Dr.Ellis	Chosen by 1 participant

Table 5 - User's preferences of AI characters in BioVRium.

Anca was the most chosen AI NPC (See Table 7). Appreciated for being “*Human-like*”, “*friendly*”, “*Knowledgeable*”, and “*professional*” (see Fig.31). **P2**, **P4**, **P8**, **P9** and **P10** mentioned Anca is balanced: neither too strict nor too casual.

P2: *“Anca - was the most human-like - the rest seem too bizarre for me - the old guy was too boring, the girl in costume had too much fake, not according to study energy”.*

P4: *“Anca, because it felt friendly and knowledgeable”,* **P8:** *“Anca. I chose her because she was a mix between Dr Ellis (too strict) and BioBear (too friendly). She seemed*

knowledgeable and welcoming”, P9: “I chose Anca, because she is clear, friendly and professional.”

It seems that this blend has supported a conducive learning environment, making participants feel comfortable yet focused on learning.

P10: *“I preferred Anca because she was the most humanized character. Also she seemed to be friendly and knowledgeable at the same time”.*

This preference aligns with research by Asne Stige (2023), which found that AI character with human-like qualities tend to be more effective in educational settings; they foster a sense of family and trust.

BioBear was chosen by participants who enjoyed a gamified learning experience.

P1: *“I had chosen the BioBear as my AI NPC, the reason for that was because it was something unique to learn microbiology from a cartoon figure”.*

P5 and **P7** mentioned that BioBear’s unique, game-like appearance helped them feel as they were in a game, rather than in a learning experience;

P5: *“Biobear, because it feels like a game rather than a lecture.”*, **P7:** *“Biobear, because I like videogames and it seemed friendly and funny, which would make the sessions less boring.”.*

Dr. Ellis, was **P3**’s choice for the AI-enhanced BioVRium. They appreciated the character’s direct and strict approach:

“Dr Elliss because he wanted to give me straight facts without compassion”.

This preference underscores research by Zhang (2017), who suggests that some learners prefer an authoritative style in educational settings. With further research, this could explore whether there’s a larger segment of users that prioritize a more traditional character to their learning style.

When analysing the participant’s feedback, it is clear that the vast majority preferred the most humanized character, which in their opinion, was Anca. However, Anca, compared to Dr.Ellis and BioBear, has not been equipped with a human voice. In fact, Anca’s voice is robotic, monotone and without any particularities. This could suggest that users might appreciate more certain human elements, such as physical aspect, rather than a humanized voice. It could be argued that this might be the result of years of AI voice-based assistants deeply integrated in the modern life, such as Siri, Alexa, etc...

This is another area that would need to be further explored and could provide meaningful insights.

Chapter 6.3 and Chapter 4, not only analyse and compare the different AI characters, but also the user’s preferences and motivations behind their choices, thus addressing directly **RQ3**.

6.4 Usability and User Experience

To determine whether adding AI to BioVRium improves usability and user experience compared to the version without AI, 2 UX forms were employed, following the System Usability Scale (SUS). Across most questions, the AI-enhanced version of the platform performed slightly better than the VR-only version. This indicates the AI component could contribute to fostering a use-centred platform (see Table 6).

10 Likert-scale questions, producing a score from 1 (lowest) to 5 (highest), aimed at determining a series of UX metrics.

Question	System Usability Scale for BioVRium with AI (MEDIAN)	System Usability Scale for BioVRium without AI (MEDIAN)
I think that I would like to use this system frequently	4	3
I found the system unnecessarily complex.	1	1
I thought the system was easy to use.	4.5	4
I think that I would need the support of a technical person to be able to use this system.	1	1
I found the various functions in this system were well integrated.	5	5
I thought there was too much inconsistency in this system.	1	1
I thought there was too much inconsistency in this system.	4	4
I found the system very inconvenient to use.	1	1
I felt very confident using the system	4.5	4
I needed to learn a lot of things before I could get going with this system	1	1

Table 6 - Table 6 - Median scores of SUS form for both versions of BioVRium

Notably, two questions showed significant differences in responses:

Frequency of use – “I think that I would like to use this system frequently” received higher scores compared to the rest of the questions; Analysing the UX form for the version with AI, I found that 7 out of 10 people selected “4”, meanwhile 2 people selected “5” (see Fig. 33). This suggests that the interactive and supportive features provided by the AI made the system more appealing for repeated use, enhancing the perceived value and engagement of the educational platform.

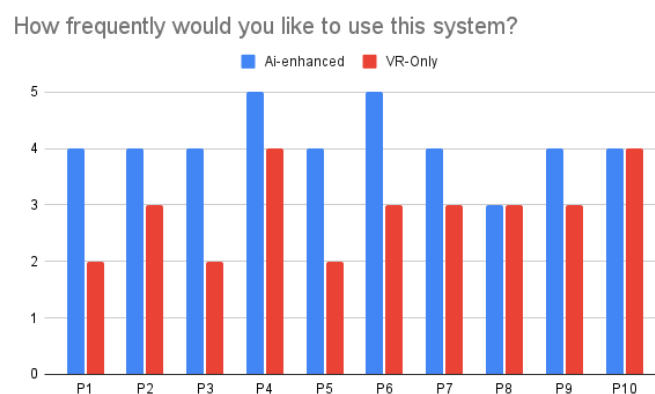


Figure 33 - Score of SUS question, comparing AI-enhanced version to VR-only.

User Confidence – “I felt very confident using the system”, was also higher in the AI-enhanced version (See Fig. 34). Participants expressed greater comfort and assurance when navigating and interacting within the VR environment, likely due to the guidance and immediate feedback offered by the AI characters.

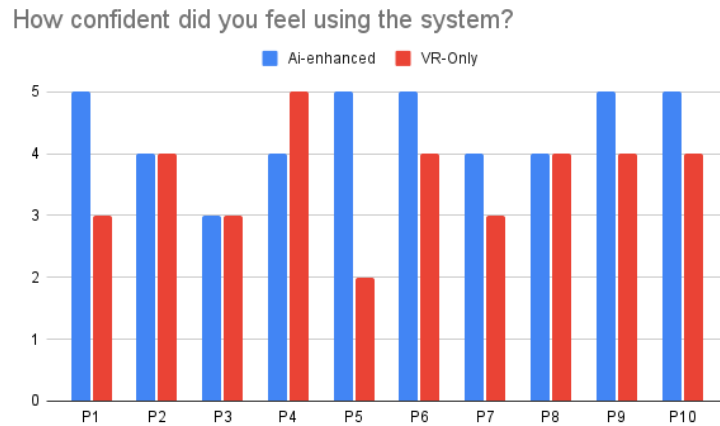


Figure 34 - Score of SUS question, comparing AI-enhanced version to VR-Only.

More data was collected from the post-assessment survey, particularly regarding interactions with AI characters. When participants were asked “On a scale of 1 to 10, how easy was it to interact with the AI character”, the score was **Median = 4** (see Fig. 35).

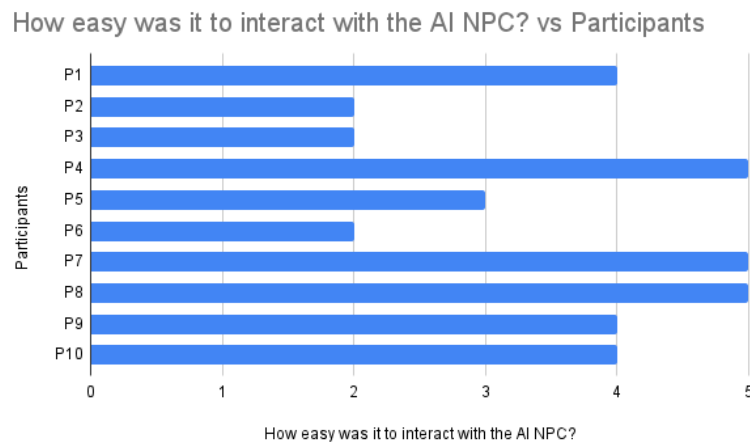


Figure 35 - Participant's insights about AI component interation.

Section 7.4 explores usability and user experience, relevant to **RQ1**. Furthermore, insights about the usability of the AI-enhanced platform and the user experience can be discussed in the context of **RQ2**.

7 Conclusion

7.1 Comparison with Literature Review

In this project, it has been demonstrated the benefits of integrating generative Artificial Intelligence with Virtual Reality by employing 2 studies, one with 31 participants to gather user requirements, and a cross-over study with 10 participants to test BioVRium's efficacy as a learning platform and its usability levels. Not only this study was *statistically significant*, proving that AI-enhanced VR platforms lead to higher engagement and improved usability, but also to better learning outcomes compared to VR-only platforms. This aligns with the work of Bermejo (2023), who highlighted the power of AI in creating learner-centric environments.

The results of this research are consistently in line with the body of literature review, which highlighted the benefits of Virtual Reality and Artificial intelligence in educational settings.

7.2 Reflection on Findings

In Chapter 6, empirical evidence has been provided to support the hypothesis that the inclusion of generative AI in a VR educational platform improves learning outcomes and usability levels. For instance, participants performed better when testing the AI-enhanced version of BioVRium, as shown in the Mann-Whitney U-Test, thereby addressing *Research Question 1 – How does the inclusion of AI in a VR platform impact the learning outcomes and usability levels of students compared to a non-AI VR approach?* –

On the same note, this research highlighted how the AI component and the presence of 3D interactive elements, can influence students' engagement and perceived learning effectiveness, which itself is a factor contributing to motivation and adherence (Asne Stige, Efraxia D Zamani, 2023). Participants who reported positive interactions with their AI guide, generally showed an improvement in engagement and test scores. This not only highlights that if users are instructed on how to interact with the AI character, and it is well integrated with the platform, it helps learners achieve better results, but it also addresses *Research Question 2 – How do the interactive and AI-enhanced features of the VR platform influence students' engagement and perceived learning effectiveness in studying microbiology?* –

Ultimately, the study also explored user preferences among the 3 available AI characters. The majority of participants preferred Anca, the character that in their opinion, is the most humanized. However, Anca was the only character with a non-human voice. This suggests that users might appreciate certain human elements, such as physical aspect, rather than a humanized voice. Hence, it directly addresses *Research Question 3 – What are initial user impressions of engaging with AI characters in a VR educational setting, and which AI character do participants prefer? Why do they prefer that character?* –

7.3 Limitations

7.3.1 Self-Reported Data

While the study measured learning outcomes and user engagement across a substantial number of surveys, the reliance on self-reported data introduces inherent limitations. Self-reported measures are valuable for gaining insights, but they can be subject to biases, such as memory recall inaccuracies. Future studies might take into consideration removing self-reported data.

7.3.2 Participants and time limitations

The participant sample size, their background and the study's duration present another set of limitations. Study 2 involved 10 participants, therefore the statistical power of this study is reduced. Additionally, while the participant group was diverse in terms of academic backgrounds, the lack of microbiology knowledge in participants might undermine the relevance of this study. The limited timeframe of 2 weeks might not adequately capture long-term learning retention or the sustained use of the VR platform. Future studies should consider employing an increased number of participants from educational settings related to microbiology and consider longer-term studies.

7.3.3 Convai Plug-in

The AI component produced some technical and performance issues. Even if the issue is related to CONVAI, such as response lags, change of voice, inaccuracy, etc... this could have potentially altered the results related to the usability and educational effectiveness of the AI component.

7.4 Further Research

There are several directions for future research to build upon the findings of this study. **Longitudinal studies** should be implemented to strengthen the statistical power of these findings and to investigate the long-term impacts of the AI component regarding retention and adherence.

Multilingual support is a capability of the AI NPC discovered during the development phase of BioVRium, however, due to time limits, it was not explored in this research. Exploring this functionality, could make the platform more inclusive and accommodating, thereby, broadening its potential impact.

Digital logbook that tracks and summarizes the microorganisms studied by the user could be a valuable addition to this platform. It would offer a personalized learning journey and enhance content retention by providing users with an overview of their progress.

Visual depictions of microbial disease impacts on human health are another functionality that could add more context and practicality, making the learning process more relevant and efficient.

Despite the limitations encountered during this study, the findings were **statistically significant** and should be further investigated. Incorporating generative AI in immersive Virtual Reality platforms for education can lead to higher engagement, improved usability and academic performance. However, the full potential of generative AI remains to be explored.

While it is just one piece of a larger puzzle, the findings provide valuable insights towards a future where learning is not confined to textbooks, but is immersive, engaging and personalized.

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Appendix

Appendix A: Flyer, Consent Form, Research Information Form, Debrief Statement,



1.

Consent Form

I have thoroughly read and understood all the details about the study mentioned above, and I have received a hard copy for my records.	YES	NO
The purpose and procedures of this research have been clearly explained to me, and I have had the opportunity to address any questions or concerns.	YES	NO
I am aware that my participation in this study is confidential and all data I provide will be treated with full confidentiality.	YES	NO
I understand that if published, it will not be identifiable as mine. Only the research team will have access to the data I provide, and I am informed about its future use once the study is completed.	YES	NO
I agree that my (anonymised) information can be quoted in research outputs.	YES	NO
I agree that the researcher may contact me to follow up on any findings or information that I provide them.	YES	NO
I give permission for the (anonymised) information I provide to be deposited in a data archive so that it may be used for future research.	YES	NO
I consent to having my voice recorded during the study.	YES	NO

Participant's Name (Print): _____

Participant's Signature: _____

Date: _____

2.

Research Information Form

Title of Study: Immersive Study of Microscopic Organisms: A Pathway to Interactive Learning

Researcher: Anca-Gabriela Craciunica

Supervisor: Dr. Stephen Uzor

Introduction

This study aims to explore how students can benefit from immersive technologies to study abstract concepts of biology, more specifically microorganisms. I am developing a VR (Virtual Reality) platform that includes 3D models and AI to enhance the learning experience.

What will happen

If you agree to take part in this study, you will be asked to participate in a learning session where you will use the VR platform and the VR platform with AI to learn about a specific topic related to microscopic organisms. The session will last approximately 60 minutes.

Testing

After the learning sessions, you will be asked to complete tests related to the topic you learned about. This test is designed to measure your understanding and retention of the material.

Survey

You will be asked to complete a survey about your experience. The survey includes open-ended questions about your interactions with the learning material and your overall experience.

Your participation in this study will last approximately 1 hour.

Location

The study will be carried out in a lecture room at Ealing Campus, St Mary's Rd, London W5 5RF.

Confidentiality

Your personal information and responses will be confidential, and we will not share them with anyone outside the research team. Once the study is completed, all the participant data will be destroyed.

Right to Withdraw

Your participation in this study is entirely voluntary, and you have the right to withdraw at any time without any consequences. Please feel free to ask me any questions. If you are happy to continue you will be asked to sign a consent form prior to your participation. Please retain this form for reference.

Contact Information

If you have any questions or concerns about this study, please feel free to contact the study's supervisor, Dr. Stephen Uzor (stephen.uzor@uwl.ac.uk).

3.

Debriefing Statement

Thank you for your participation in this study. Your time and contribution are highly appreciated.

The purpose of this study was to investigate the potential of an immersive learning experience for microorganisms by developing a VR platform that includes 3D models and AI. The goal is to facilitate the learning process of such complex concepts and make them more accessible and engaging, especially for students who may struggle with traditional learning methods.

During your participation, you were asked to engage with either the VR platform or the VR platform with AI to learn about a specific topic related to microscopic organisms. You were then tested on your understanding and retention of the material. You also completed a survey about your experience.

The data collected from this study will be used to evaluate the effectiveness of the VR platform in enhancing the understanding and retention of information about microscopic organisms.

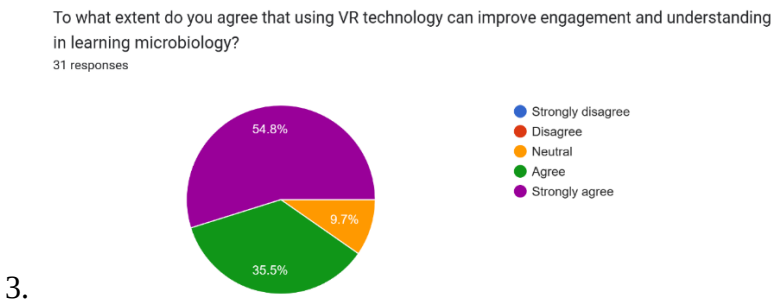
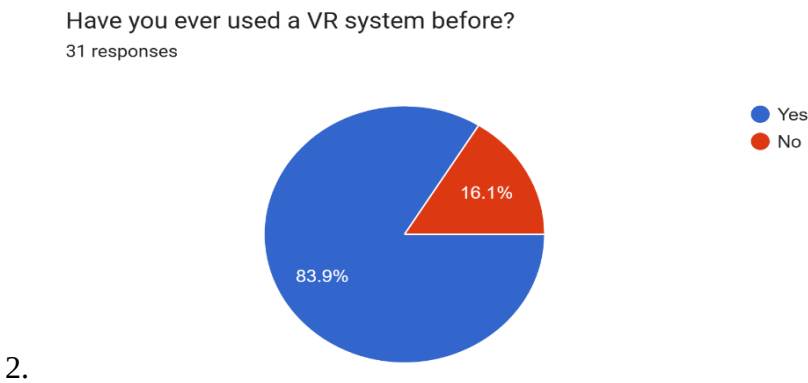
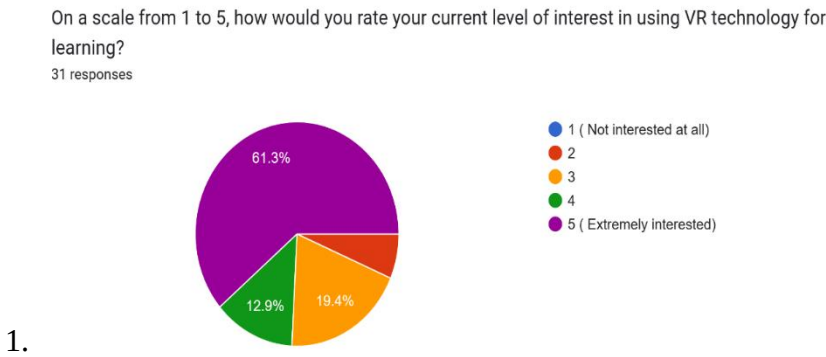
Please remember that any data about your participation in this study will remain confidential. If published, it will not be identifiable as yours.

If you have any questions or concerns about this study, please feel free to contact the study's supervisor, Dr. Stephen Uzor (stephen.uzor@uwl.ac.uk).

4. Thank you once again for your valuable contribution to this research.

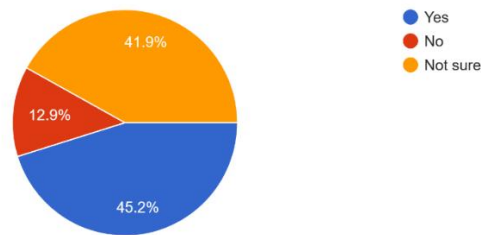
Appendix B: User Requirements Survey, Pre- and Pos-Assessment Survey

User Requirements Survey.



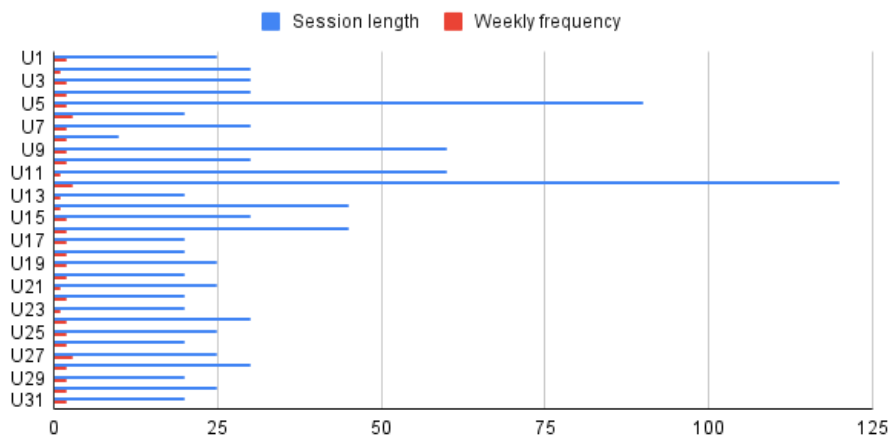
Should the VR platform be usable across multiple devices (e.g., stand-alone VR headsets, PC-tethered headsets, mobile devices)?

31 responses



4.

In your opinion, what is the ideal length of a single VR session for studying microbiology, and how often should these sessions occur to maximise learning? In your opinion, what is the ideal length of a single VR session for studying microbiology, and how often should these sessions occur to maximize



5.

What suggestions do you have for making VR a more engaging and effective tool for all students, in the study of microbiology?

31 responses

Sandbox environment for exploration or experimentation.

I'm not really sure. There's a VR game named out of scale thats educational and deals with similar content so maybe you could get ideas from that game though

Make the knowledge required for the core gameplay, make the gameplay enticing to players who want to apply microbiology to the gameplay, that way the "tests" are whether your faction or creature survives better because of further microbiology knowledge

No long lectures and more interactive things

.

If you make it like a game, people will be more inclined to learn from it. Either way though, I'm excited to see where this goes! VR is a powerful tool.

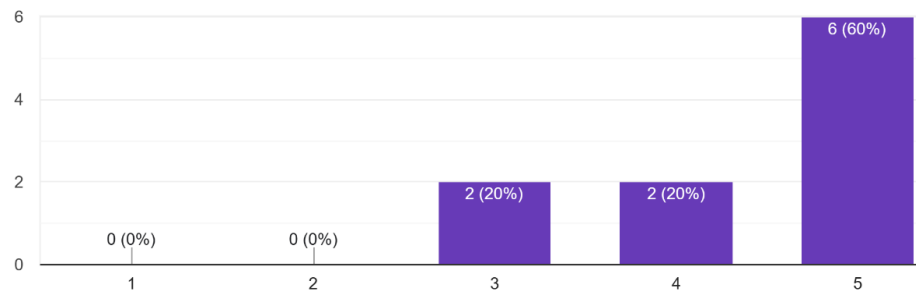
Make is more immersive, well animated.

6.

Pre-assessment survey.

How effective do you think learning microbiology in a VR environment will be?

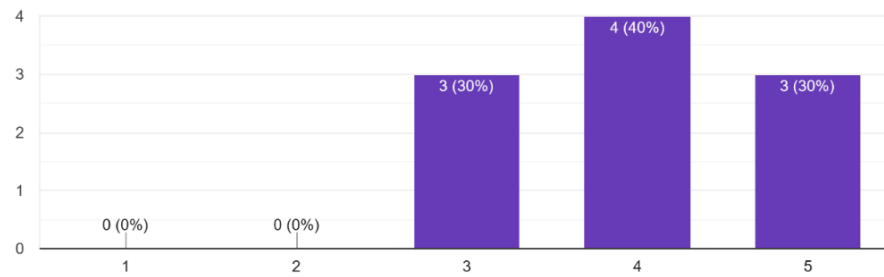
10 responses



1.

On a scale of 1 (not comfortable at all) to 5 (extremely comfortable), how comfortable are you with interacting with AI?

10 responses

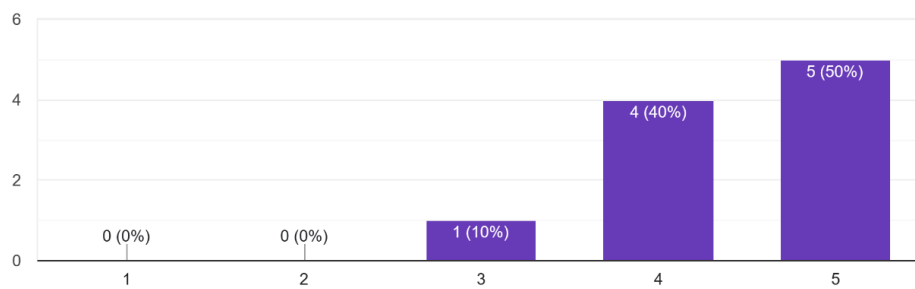


2.

Post-Assessment Survey

PART 1: SOFTWARE WITHOUT AI How easy was it to navigate the VR environment?

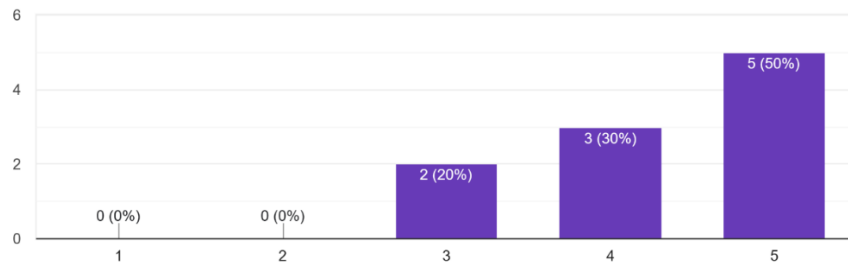
10 responses



1.

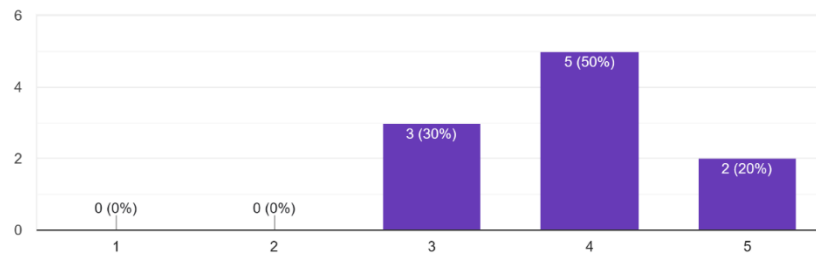
How satisfied were you with the VR platform?

10 responses



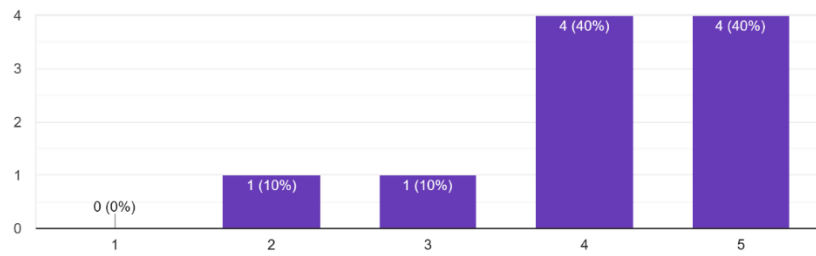
2.

How much do you feel your knowledge of microbiology improved after using the VR platform?
10 responses



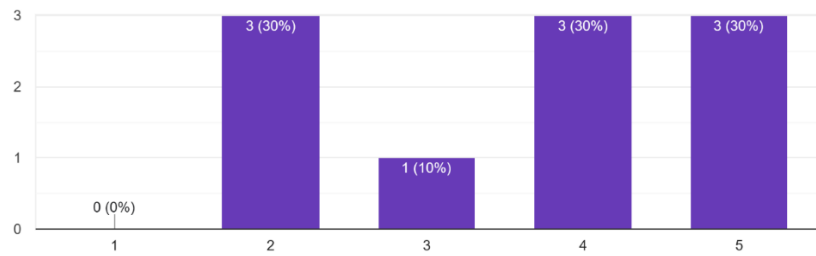
1.

PART 2: VR WITH AI How easy was it to navigate the VR environment? (Rate on a scale of 1-5)
10 responses



2.

How easy was it to interact with the AI NPC?
10 responses



3.

Which AI NPC did you prefer/choose and why?
10 responses

I had chosen the BioBear as my AI NPC, the reason for that was because it was something unique to learn microbiology from a cartoon figure.

Anca - was the most human-like - the rest seem too bizarre for me - the old guy was too boring, the girl in costume had too much fake, not according to study energy

Dr Elliss because he wanted to give me straight facts without compassion

Anca, because it felt friendly and knowledgeable

Biobear, because it feels like a game rather than a lecture.

A. Anca because of her uniform and her voice was clear and exciting

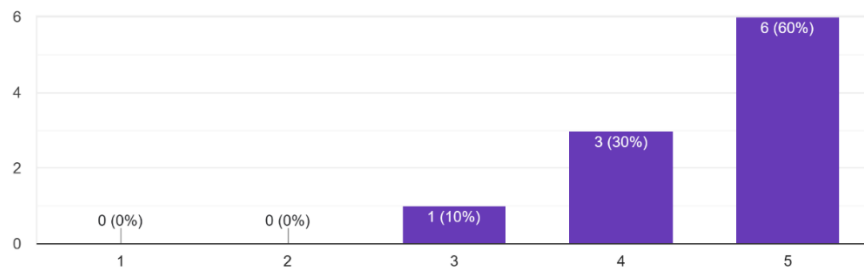
Biobear, because I like videogames and it seemed friendly and funny, which would make the sessions less boring.

Anca. I chose her because she was a mix between Dr Elliss (too strict) and BioBear (too friendly). She

4.

How satisfied were you with the VR platform?

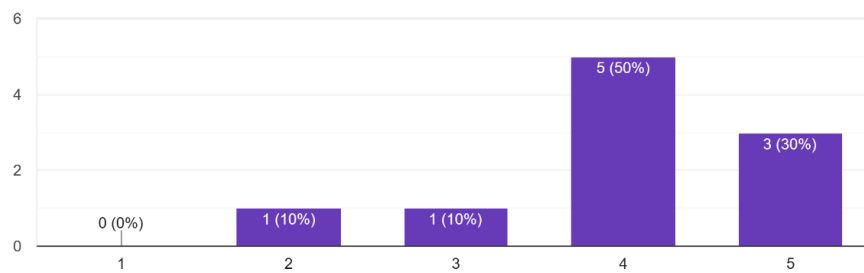
10 responses



5.

How much do you feel your knowledge of microbiology improved after using the VR platform with the AI component?

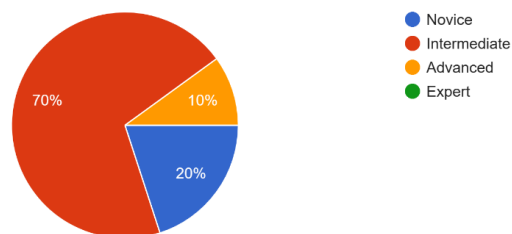
10 responses



6.

PART 3: OVERALL ASSESSMENT After using the VR platform, how would you rate your knowledge about microbiology?

10 responses



7.

Which version of the software did you find more effective for learning about microbiology(with AI without AI)? Please explain why.

10 responses

I enjoyed learning with AI as it brought fun to my study while interacting with the AI.

Short-term the version without AI is more effective, however, if I would be able to cooperate with AI effectively, I believe it could have a positive impact on my general knowledge, as it had way more to offer and explain.

Without AI ,interacting with the AI was a bit of a challenge having to go back to where it is to communicate

With AI. It was helpful to have a guide in a new environment like VR platforms are, in the same time the AI lets you explore alone and at the same time it's there to help you when you face any difficult situation.

The one with AI was more effective, because I could always ask for clarification or further help in real time. However, the AI character was very laggy, does not reply instantly and sometimes does not understand what I am saying;. Overall, I think this application if it was fully developed it could help grow my knowledge of microbiology, as of right now my knowledge remains the same level because it was a relatively short experience and even if I feel I learned a couple of interesting facts, I would not consider myself anything more than a novice.

8.

Appendix C: UX FORM

This is a standard usability survey and it was administered for both version of BioVRium..

1. I think that I would like to use this system frequently.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

2. I found the system unnecessarily complex.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

3. I thought the system was easy to use.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

4. I think that I would need the support of a technical person to be able to use this system.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

5. I found the various functions in this system were well integrated.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

6. I thought there was too much inconsistency in this system.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

7. I would imagine that most people would learn to use this system very quickly.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

8. I found the system very cumbersome to use.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

9. I felt very confident using the system.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

10. I needed to learn a lot of things before I could get going with this system.

1. Strongly Disagree	2.	3.	4.	5. Strongly Agree
☐	☐	☐	☐	☐

1.

This table shows the SUS results, comparing the scores of both versions.

Question	System Usability Scale for BioVRium with AI (MEDIAN)	System Usability Scale for BioVRium without AI (MEDIAN)
I think that I would like to use this system frequently	4	3
I found the system unnecessarily complex.	1	1
I thought the system was easy to use.	4.5	4
I think that I would need the support of a technical person to be able to use this system.	1	1
I found the various functions in this system were well integrated.	5	5
I thought there was too much inconsistency in this system.	1	1
I thought there was too much inconsistency in this system.	4	4
I found the system very inconvenient to use.	1	1
I felt very confident using the system	4.5	4
I needed to learn a lot of things before I could get going with this system	1	1

Appendix D: Documentation

This is the virus test that participants had to take when completing Level 3 (Viruses).

Virus Quiz

1) Adenovirus Question: "How does adenovirus mainly spread?" A) Through water B) By close personal contact

Correct Answer: B

2) Coronavirus Question: "What gives the coronavirus its name?" A) Its rod shape B) Its crown-like spikes

Correct Answer: B

3) Ebolavirus Question: "What is a common symptom of Ebola virus disease?" A) Sneezing B) Hemorrhaging

Correct Answer: B

4) Hepatovirus A Question: "What disease does Hepatovirus A cause?" A) Hepatitis A B) Hepatitis B

Correct Answer: A

5) HIV Question: "What system does HIV attack in the body?" A) Digestive system B) Immune system

Correct Answer: B

6) Influenza Virus Question: "What kind of RNA does the influenza virus have?" A) Single-stranded B) Double-stranded

Correct Answer: A

7) Rabies Lyssavirus Question: "How is rabies typically transmitted?" A) Through the air B) From animal bites

Correct Answer: B

Varicella Zoster Virus Question: "What does the varicella zoster virus cause in children?" A) Chickenpox B) Cold sores

1. Correct Answer: A

This is the bacteria test that participants had to take when completing Level 4 (Bacteria)

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Bacteria Quiz

1)Bacillus anthracis Question: "Which environment is Bacillus anthracis naturally found in?" A) Soil B) Air
Correct Answer: A

2)Bifidobacterium Question: "What is Bifidobacterium commonly used as?" A) Pathogen B) Probiotic
Correct Answer: B

3)Haemophilus influenzae Question: "Despite its name, does Haemophilus influenzae cause the flu?" A) Yes B) No
Correct Answer: B

4)Meningococcus (Neisseria meningitidis) Question: "What serious condition can Meningococcus cause?" A) Meningitis B)
Influenza
Correct Answer: A

5)Mycobacterium tuberculosis Question: "What disease is caused by Mycobacterium tuberculosis?" A) Tuberculosis B)
Bronchitis
Correct Answer: A

6)Salmonella Question: "How do people typically get infected with Salmonella?" A) Contaminated food B) Airborne droplets
Correct Answer: A

7)Staphylococcus Question: "What shape are Staphylococcus bacteria?" A) Rod-shaped B) Sphere-shaped
Correct Answer: B

8)Treponema pallidum Question: "What does Treponema pallidum cause?" A) Syphilis B) Tuberculosis
Correct Answer: A
```

2.

This is the structure provided to the AI characters to provide a guide to the users when they interact in Level 1.

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Tutorial

Welcome to BioVRium, a VR experience designed to immerse you in the fascinating world of microbiology. I am Anca, your
virtual guide, here to assist you on this educational journey. To talk with me, simply press and hold the "A" button on
your right controller and release it when you're done speaking.

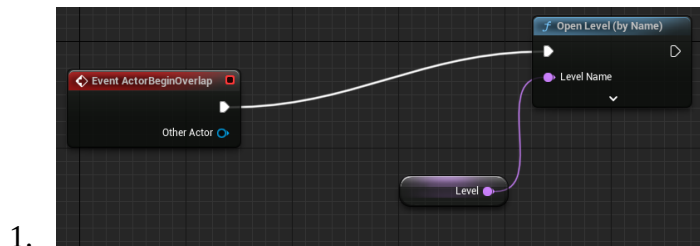
Before we begin, let's ensure your safety. Clear your play area to avoid obstacles, and remember, you can always take a
break if you feel discomfort. Our goal here is to learn about the intricate lives of microorganisms. You'll move and
explore using the thumbsticks, and interact with the environment using the trigger buttons. If you ever feel uneasy, the
teleportation feature is a comfortable way to navigate without inducing motion sickness.

Lastly, please confirm your understanding of these instructions and give a quick nod or a verbal "yes," and we'll embark on
this microscopic adventure together. Our adventure starts at the microscope on the table in front of you. Approach it and
use your controller to interact with it--this will transport us into the microscopic realm where learning comes to life.
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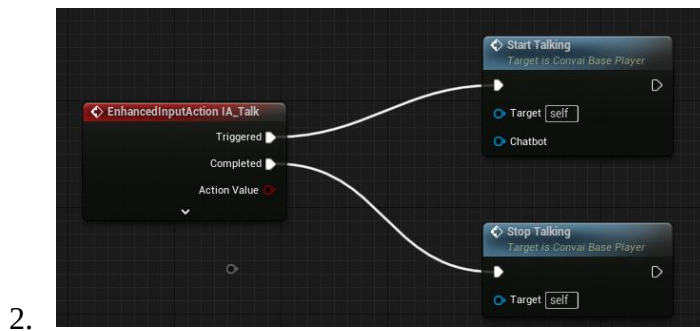
3.

Appendix E: Blueprints

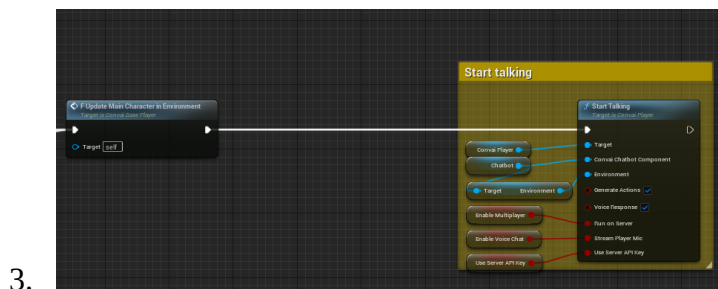
This is the blueprint used to teleport users from one level to the other. It has been used for the microscope to access level 2, and for the portals to access level 3 and 4.



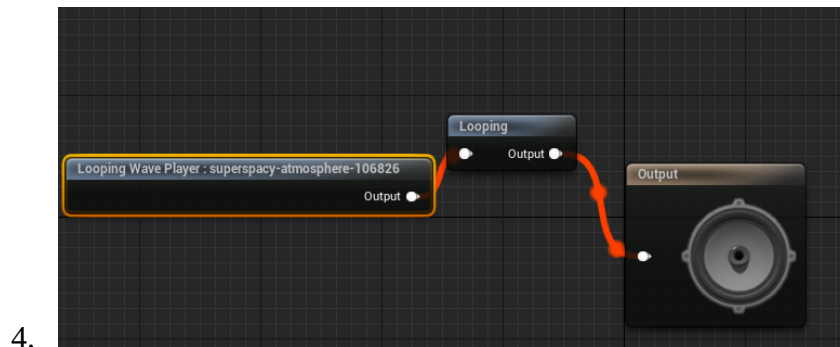
The function of this blueprint is to enable users to talk with the AI guide by pressing the button “A” on the right controller. The data set IA_Talk was created for this purpose.



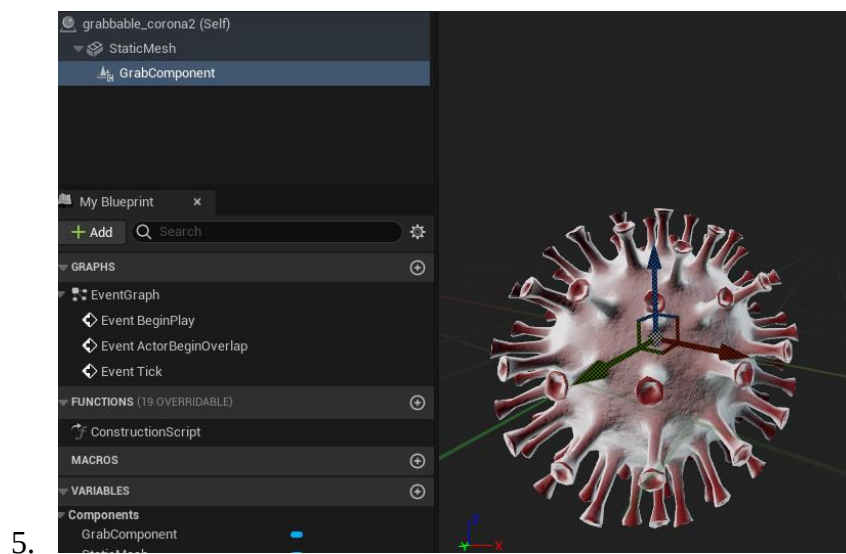
The function of this blueprint is to identify and connect the main character in the environment (in our case, the CONVAI) to the CONVAI Chatbot component and to enable the talking function.



The function of this blueprint is to enable the wave to be on a loop for each level, hence adding background music.



This shows the structure of a blueprint created for each microorganism.



APPENDIX F: **BIOVRIMUM**

To provide a comprehensive view of BioVRium, a presentation is available on youtube:
[BioVRium- 21507346- UWL \(youtube.com\)](#)